# Limited Conservation Benefits From the First Marine Protected Area on the High Seas

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**Abstract**

Can Large-Scale Marine Protected Areas be effective on the high seas? We address this question by evaluating the world's first large-scale fishery closure implemented in areas beyond national jurisdiction. In the Western and Central Pacific Ocean, two large high-seas pockets were closed to industrial purse-seine fishing in 2010 to reduce bycatch of juvenile bigeye tuna, but industrial longline fishing was allowed to continue. Nearly two decades later, the empirical consequences of this voluntary non-binding policy remain unknown. Using publicly available catch and effort data, we find that the closure essentially eliminated purse-seine fishing within the high-seas pockets. We also find that longline vessels operating within the closed areas experienced a weakly significant 19.6% increase in the number of bigeye tuna caught per 1,000 hooks. But we find no evidence that the closure generated a spillover benefit to purse-seine vessels fishing near the boundaries of the closed areas.

**Significance statement.** The UN High Seas Treaty entered into force in January 2026, yet no empirical evidence exists on whether Marine Protected Areas can deliver conservation benefits beyond national jurisdiction. We provide this evidence by evaluating the world's first large-scale high-seas closure—two vast Pacific fishing grounds totaling over 1,000,000 $km^2$ closed to industrial purse seining since 2010. Analyzing two decades of spatially explicit catch and fishing effort data, we show that market-based (as opposed to legally binding) enforcement successfully eliminated industrial purse seine fishing from a large swath of international waters. However, the resulting conservation benefits were far more modest than those documented for comparable closures within national waters, and no spillover reached adjacent fisheries. As nations begin creating high-seas MPAs under the new UN treaty, our findings provide the first empirical benchmark for what these protections may—and may not—achieve.

# 1 Introduction

The entry into force of the UN High Seas Treaty (BBNJ Agreement) on January 17, 2026, officially established the legal framework necessary to create and manage Marine Protected Areas (MPAs) on the high seas. By definition, high seas MPAs will protect pelagic ecosystems. In this way, they will share some similarities with Large-Scale MPAs (LSMPAs; areas larger than 100,000 $km^2$ [1, 2]) implemented in Exclusive Economic Zones (EEZs). By virtue of the habitats they protect, LSMPAs most often restrict fishing effort of pelagic fisheries, such as those targeting tuna, billfish, and sharks [3]. These fisheries are also likely to be displaced by high seas MPAs.

Previous work has found that tuna fisheries operating near LSMPA boundaries may see increases in their catch rates [4, 5, 6]. For example, Hilborn et al. [6] find that the catch-per-unit-effort (CPUE) of bigeye tuna has increased by 13% outside the boundaries of Papahānaumokuākea National Marine Sanctuary in Hawai‘i. These potential spillover benefits may prompt nations to think strategically when designing LSMPAs on the high seas. For instance, some may implement LSMPAs immediately outside their Exclusive Economic Zones (EEZs) hoping to boost catch rates in their waters [7, 8]. But would this work?

There are subtle but important distinctions between Large-Scale MPAs inside EEZs and Large-Scale MPAs on the high seas. One important difference is enforcement and compliance. Not all nations have ratified the UN Convention on the Law of the Sea and it is still unclear who would be responsible for enforcing and potentially confronting illegal fishing in future high seas MPAs. A second consideration is that of differences in habitat and environmental conditions. High seas MPAs will all be located at least 200 nautical miles away from land, in some of the deepest and most remote marine ecosystems in the world. Many of the processes that influence pelagic species around islands and seamounts are either absent or very different on the high seas [9, 10, 11, 12]. Therefore, it is not clear whether results from LSMPAs within national jurisdictions will extrapolate to similarly sized MPAs on the high seas [13]. Here, we fill this gap in policy-relevant knowledge by studying the first known case of large-scale area-based marine conservation on the high seas.

We focus on the Western and Central Pacific Ocean, where industrial tuna fishing vessels use two main gears to target different tuna species. The longline fleet primarily catches albacore (*Thunnus*

*alalunga*), bigeye (*Thunnus obesus*), and yellowfin tuna (*Thunnus albacares*). The purse seine fleet mainly catches skipjack (*Katsuwonus pelamis*) and yellowfin tuna, with some bycatch of juvenile bigeye tuna associated with the use of drifting Fish Aggregating Devices [14, 15, 16].

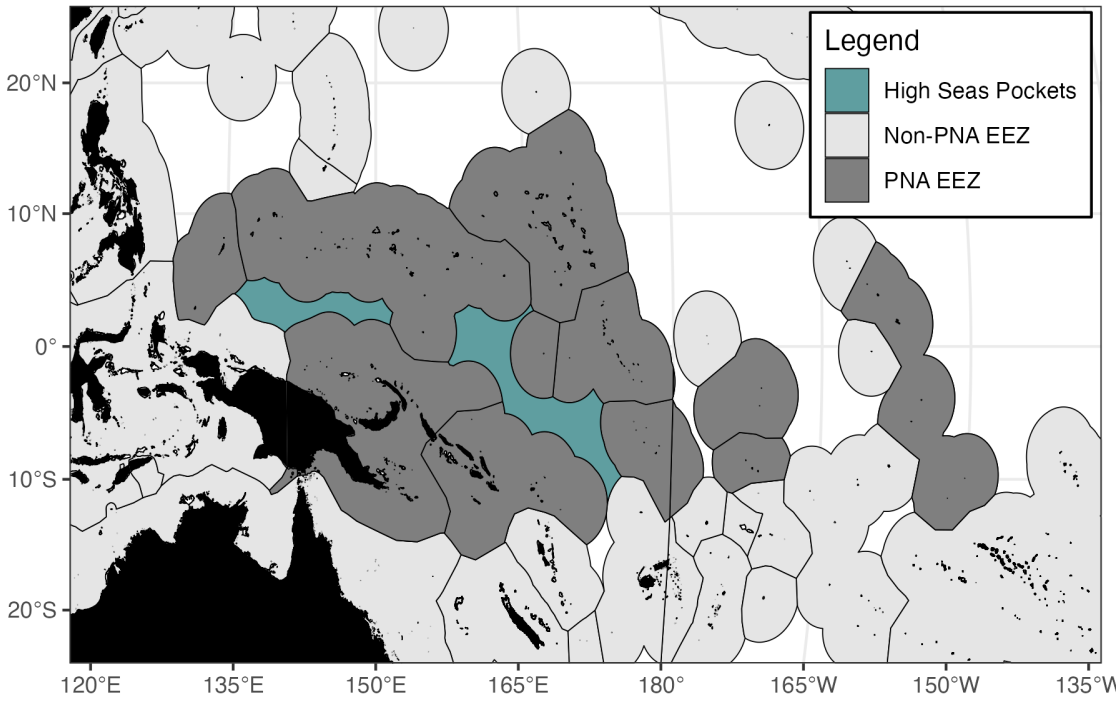



Figure 1: Map of the Western and Central Pacific Ocean. Black polygons show land. Light gray polygons show the Exclusive Economic Zones of non-PNA nations. Dark gray polygons highlight the Exclusive Economic Zones of the eight Parties to the Nauru Agreement nations and Tokelau. The two teal polygons show the High Seas Pockets—the "donut holes" formed by the surrounding Exclusive Economic Zones—where purse seine fishing has been prohibited since 2010.

The Parties to the Nauru Agreement (PNA)—a group of eight nations[1] in the region—and Tokelau jointly manage tuna fisheries in their waters using a market-based approach known as the Vessel Day Scheme (VDS; [17, 18, 19]). Bycatch of juvenile tuna by the purse seine fleet presented a major problem to the longline fleet [14, 15, 16], particularly in the two high seas pockets formed between the PNA's Exclusive Economic Zones (EEZs; Figure 1). When the spatial overlap between target and bycatch species is severe, MPAs may be a useful tool to protect the "weak stock" [20], here juvenile bigeye. In an effort to reduce the purse-seine fleet's bycatch of

[1]PNA members include: Federated States of Micronesia, Kiribati, Nauru, Palau, Papua New Guinea, Republic of the Marshall Islands, Solomon Islands, and Tuvalu.

juvenile bigeye tuna—an externality affecting longline vessels targeting adult bigeye tuna—purse seine fishing was prohibited in the high seas pockets starting in 2010 [21, 22].[2] These pockets are roughly 360,000 km$^2$ and 780,000 km$^2$, so they are very much in line with the size of LSMPAs in the region, like the Palau National Marine Sanctuary (475,077 km$^2$) or the now-reopened Phoenix Islands Protected Area (408,422 km$^2$). In closing the high seas pockets to industrial tuna purse seine fisheries, this intervention effectively created the first partially protected *de facto* large-scale MPA in areas beyond national jurisdiction. How was the prohibition on purse seine fishing achieved before the UN High Seas Treaty even existed? The PNA used a carrot-and-stick approach: vessels that adhered to the closure were allowed to fish in the lucrative waters of the PNA countries under the VDS program; vessels that ignored the purse seine ban were excluded from the VDS program.

There are only a few examples of high seas closures implemented before the adoption of the high seas treaty, and their potential effectiveness remains contested or unevaluated [13]. In particular, previous work on the PNA high seas pocket has predicted that the benefits to tuna populations would be limited. Sibert et al. [34] used hind-cast simulations to estimate the impact that the high seas pocket closures would have on bigeye and skipjack tuna populations, finding that they would have negligible stock-wide effects (0.1-0.8% increase) but potential for spillover (10-25% localized increase), depending on modeling assumptions. However, nearly two decades later, the actual effectiveness of this spatial intervention remains untested.

Our objective is to test for the effectiveness of this unique policy intervention: the first large-scale spatial closure on the high seas. We compile a multi-decadal (2000-2020) and spatially explicit dataset on catch and effort by purse seine and longline fisheries using publicly available information from the Western and Central Pacific Fisheries Commission (WCPFC; see section 3). We present our analysis in three parts. First, we ask whether the intervention actually reduced purse seine fishing in the high seas pockets. Second, we test whether the reduction of purse seine fishing effort benefited the longline fleet that still fishes in the pockets (as originally hoped for). Third, we test whether the high seas pockets, acting as a *de facto* marine protected area, provided a spillover benefit to tuna fisheries fishing outside but near the high seas boundaries (an unintended benefit

[2]The WCPFC introduced an exception for the Philippine artisanal purse seine fleet, which has allowed limited fishing by a small number of vessels in the western pocket since 2012 [23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33].

of this policy, but a pattern observed in other settings). We answer these questions following a "pre-registered" study design available on the Open Science Framework platform [35]. As we will show, the intervention was successful at reducing purse seine fishing from the high seas pockets, but we only observe weak evidence that this led to a benefit to the longline fleet (via increased catch rates in bigeye), and no evidence of a spillover benefit to any fleet. Our findings can help inform the design of, and manage expectations around, proposed LSMPAs on the high seas.

# 2 Results and Discussion

## 2.1 Reducing fishing effort by purse seiners

The closures sought to reduce purse seine fishing effort from two high seas pockets. Consequently, we begin by testing for changes in purse seine fishing effort in the high seas pockets. A visual inspection of trends in the data reported by the WCPFC shows that, in the 10 years leading to the closure, total annual fishing effort by purse seiners fishing in the high seas pockets was around 2,187 $\pm$ 770 days (mean $\pm$ standard deviation) or 1,766 $\pm$ 634 sets (Figure 2A-B). After the closure, fishing effort in the pockets dropped to zero starting in 2010, except for a few events recorded in 2013 and 2019.

To conclusively claim that the reduction in fishing effort is due to the policy change, we must compare trends in fishing effort inside the high seas pockets to trends in fishing effort in nearby comparable areas. These "control" areas must be subject to the same environmental conditions as well as fishery management rules, thus providing a counterfactual of expected fishing effort had the closure not been implemented. With this in mind, we compare all $1^\circ \times 1^\circ$ grid cells in the high seas pockets (N = 40) to other high seas grid cells that fall under the WCPFC convention area and lie between 20°N and 20°S (N = 151; see Figure A4). We estimate a two-way fixed effects regression model that accounts for differences across grid cells and years to test for a causal effect of the closure (see Methods in section 3.4.1 for more details).

In line with the patterns in Figure 2, we find that the closure dramatically reduced fishing effort in the high seas pockets, with an estimated annual reduction of 59.47 days or 50.93 sets per grid

cell per year ($p < 0.01$), relative to trends observed for other high-seas areas in the region (Table 1). We also estimate a model with dynamic treatment effects that measures the change in fishing effort by year. These event-study regressions reveal two important findings. First, that pre-treatment estimates are small relative to the post-treatment effect and show no systematic trend, though a few individual years differ statistically from zero. And second, that all years since 2010 had significantly less fishing effort than what would have been expected, had the closure never been implemented ($p < 0.01$; Figure 2C-D). Results for other specifications are provided in Supplementary Materials section B.1.

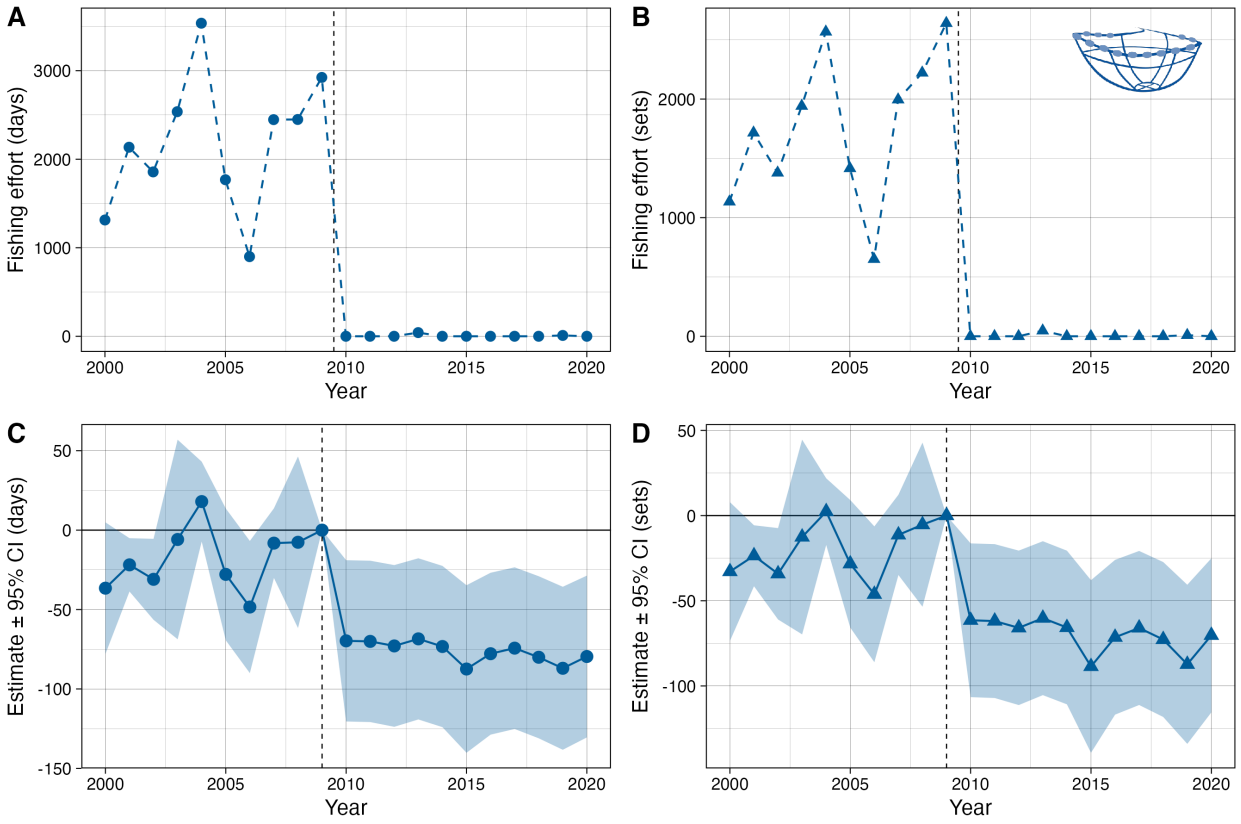



Figure 2: Change in purse seine fishing effort in the high seas pockets (combined). Panels A and B show annual total fishing effort measured in days and number of sets, respectively, across all grid cells in the high seas pockets. Panels C and D show event-study plots for the change in fishing effort in the high seas pockets, relative to fishing effort in comparable high seas regions in the WCPFC convention area.

Table 1: Coefficient estimates for the change in fishing effort inside the high seas pockets after the closure, relative to the change in fishing effort observed for other comparable high seas areas in the WCPFC convention area.

| | Effort (days) | Effort (sets) |
|---|---|---|
| Post x Treated | $-59.470^{***}$ | $-50.933^{***}$ |
| | (13.763) | (10.608) |
| $\bar{Y}_{pre}$ | 54.670 | 44.145 |
| Num.Obs. | 4011 | 4011 |
| $R^2$ Adj. | 0.388 | 0.301 |

* p < 0.1, ** p < 0.05, *** p < 0.01
The unit of observation is a grid cell in a given year. All model specifications include fixed effects by year and grid cell. Numbers in parentheses are Conley standard errors using a 220km radius. $\bar{Y}_{pre}$ indicates the mean of each outcome variable over the pre-closure period.

## 2.2 Benefits to the longline fishery

In the 10 years before closure (2000-2009), tuna purse seiners caught an average of 48,500 mt of skipjack tuna, 13,800 mt of yellowfin tuna, and 3,200 metric tons (mt) of bigeye tuna per year inside the two high seas pockets (Figure A1). To put 3,200 metric tons in context, the bigeye tuna catch limit for the US-flagged longline fleet for the entire Western and Central Pacific in 2023 was 3,554 metric tons [36]. Our results show that purse seine fishing effort was significantly reduced in the high seas pockets in 2010. It follows, then, that so were catch and bycatch associated with this effort. Recall that the longline fleet has always been allowed to operate in the high seas pockets, so now we ask: Did reducing purse seine fishing mortality in the high seas pockets benefit the longline fleet?

We focus on bigeye tuna because the closure was motivated by high bycatch rates of juvenile bigeye in the purse seine fishery [14, 34, 15, 16]. We measure benefits to the longline fleet using CPUE of bigeye tuna. Specifically, we measure CPUE as the number of fish caught for every thousand hooks and also metric tons caught for every thousand hooks, which can respectively be

understood as imperfect proxies for local abundance and biomass [37, 38]. Time series of both measures of bigeye CPUE from the longline fleet operating in the high seas pockets and control regions are shown in Figure 3A-B. Both measures show a steady decline in CPUE during the ten years before the exclusion of purse seiners. After the closure, the decreasing trend slows down and CPUE shows relative stability, particularly when measured in number of fish per thousand hooks (Figure 3A). Note, too, that there was a sharp increase in CPUE starting around three years after the closure, peaking at around five years, and then returning to a steady level seven years after the closure.

As before, a formal test requires that we compare trends in CPUE in the high seas pockets to an appropriate control region. In this second test, we retain all $5^\circ \times 5^\circ$ grid cells[3] that fall in the high seas pockets (our "treated" cells; N = 16) and all high seas grid cells in the WCPFC convention area that lie between 20°N and 20°S (N = 69; Figure A5). We then estimate annual changes in bigeye CPUE in high seas pocket grid cells, relative to changes observed in our counterfactual grid cells (see Methods section 3.4.2). We detect a modest increase in bigeye CPUE of 0.69 fish per one thousand hooks ($p < 0.1$), and no significant change in CPUE when measured in tons per one thousand hooks (-0.003 mt per thousand hooks; $p = 0.81$; Table 2). In terms of percentage changes, the increase in bigeye CPUE represents a 19.6% increase above the pre-treatment mean CPUE of 3.54 fish per one thousand hooks inside the pockets. The contrasting results across the two measures of CPUE suggest that fishers may have caught more fish per unit effort after 2010, but that these fish were, on average, slightly smaller (perhaps from juvenile cohorts that evaded purse seine mortality). It is also very likely that the weight measure has more measurement error because it is standard practice in longline fisheres to record every bigeye tuna caught, but not every fish is weighed individually, and weights may be approximations made by the crew or extrapolated at the point of landing. Measurement error will bias coefficient estimates towards zero, in line with what we observe here.

Our event-study plots echo these aggregate results: the annual number of bigeye tuna caught per thousand hooks exhibits a slightly positive trend since 2010, but we observe no change in bigeye

[3]Publicly available longline data are reported on a $5^\circ \times 5^\circ$ grid, while purse seine data are reported using a $1^\circ \times 1^\circ$ grid.

CPUE when measured in metric tons per thousand hooks (Figure 3C-D). The lack of a monotonic increase in bigeye CPUE inside the high seas pockets could be driven by changes in fleet dynamics, whereby fishing effort increases in response to localized increased abundance of the target species. Indeed, fishing effort seems to have increased slightly between 2010 and 2016 (Figure A2). A very similar pattern appears for yellowfin tuna, where we estimate a 39% increase in CPUE measured in numbers of fish per thousand hooks (coefficient = 0.329, $p < 0.05$; Table B6). We read this result as suggestive because it is present only in the log-transformed specification and only in numbers rather than weight. Yellowfin was also not the focal species of our pre-registered hypothesis, and is one of several species-level specifications we examine, as shown in the Supplementary Materials section B.2. Our Supplementary Materials (section C.1) also include a robustness test where we estimate the effect on each high seas pocket to account for the exemption granted to Philippines artisanal purse seine fleet [23]. The increase in bigeye CPUE remains present in both pockets when evaluated individually (Table C1).

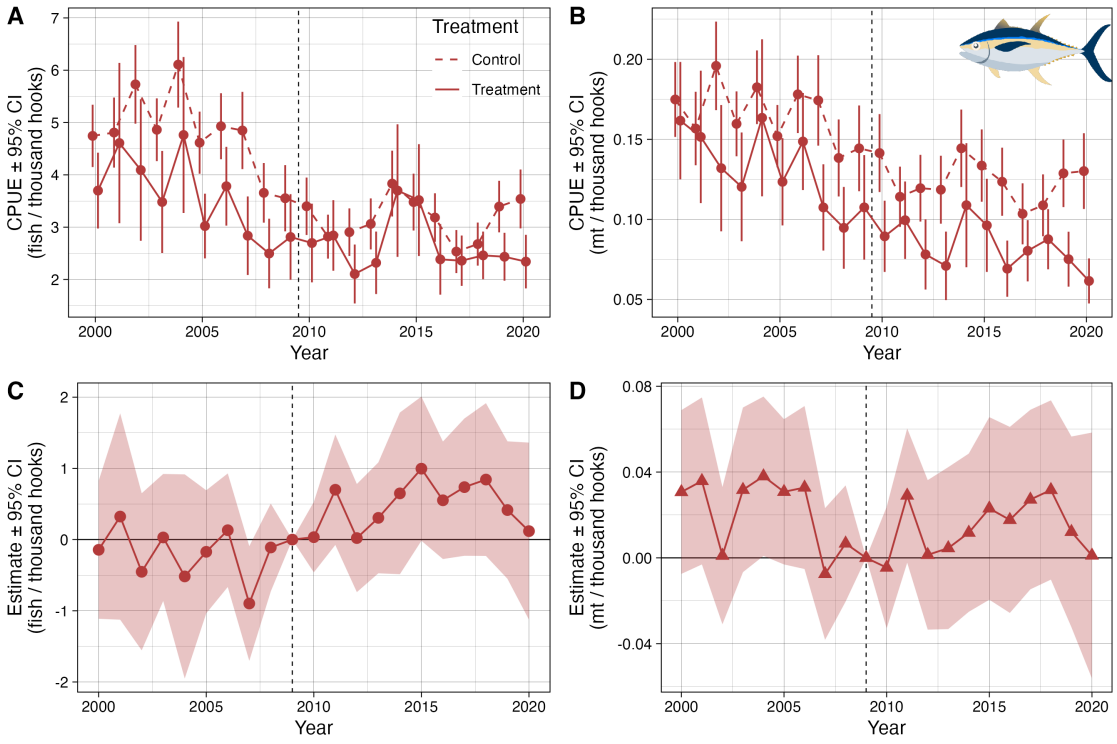



Figure 3: Change in catch-per-unit-effort (CPUE) of bigeye tuna in the longline fleet. Panels A and B show time series of mean CPUE in number of fish per 1,000 hooks and metric tons per 1000 hooks, respectively, across all grid cells in the high seas pockets (solid lines) and other tropical (20°S to 20°N) high seas grid cells in the WCPFC convention area (dashed lines). Error bar show 95% confidence intervals. Panels C and D show event-study plots for the change in bigeye tuna CPUE in the high seas pockets, relative to the change in CPUE in other tropical high seas grid cells. Ribbons show 95% confidence intervals.

Table 2: Coefficient estimates for the change in bigeye tuna CPUE in the longline fleet in the high seas pockets after the closure, relative to the change in CPUE observed for other tropical (20°S - 20°N) high seas areas in the WCPFC convention area.

| | fish / 1000 hooks | mt / 1000 hooks |
|---|---|---|
| Post x Treated | 0.697* | −0.003 |
| | (0.363) | (0.011) |
| $\bar{Y}_{pre}$ | 3.548 | 0.131 |
| Num.Obs. | 1631 | 1631 |
| $R^2$ Adj. | 0.798 | 0.829 |

* p < 0.1, ** p < 0.05, *** p < 0.01
The unit of observation is a grid cell in a given year. All model specifications include fixed effects by year and grid cell. Numbers in parentheses are Conley standard errors with a 1100km radius. $\bar{Y}_{pre}$ indicates the mean of each outcome variable over the pre-closure period.

## 2.3 Spillover benefits to the purse seine fishery

Skipjack tuna in the open ocean are mainly targeted by purse seine vessels, with the longline fishery representing less than 1% of total skipjack caught in the WCPFC convention area [39]. When the high seas pockets were closed to purse seine fishing, the region essentially became a *de facto* no-take zone for skipjack tuna. Previous work has shown that the CPUE of tuna purse seiners often, but not always, increases near large-scale, no-take MPAs after their implementation [5]. Therefore, we now ask whether the high seas pockets, by substantially reducing localized fishing mortality on skipjack, also produced spillover benefits to purse seine fisheries operating near the borders of the pockets. A direct examination of a time series of skipjack tuna CPUE in Figure 4 shows two contrasting trends. When measured in metric tons *per day*, CPUE within 100 nautical miles of the high seas pockets appears to increase after 2010, as compared to pre-closure years (Figure 4A). We observe the opposite when we measure CPUE in metric tons *per set*, which shows lower CPUE for most years after the closure (Figure 4B).

Following previous work [4, 5, 6], we formally test for changes in skipjack CPUE near the high seas pocket boundaries (within 100 nm; N = 226), relative to changes observed far from the

boundary (100–200 nm; N = 184; Figure A6). We only retain grid cells that fall within PNA countries' waters because purse seine fishing is strictly managed under the vessel-day scheme in these waters [17, 5]. Using a standard Before-After-Near-Far design, we fail to detect changes in CPUE that would be indicative of a spillover effect, with estimated effects of $-0.49$ tons per day and $-0.86$ tons per set ($p = 0.45$ and $p = 0.29$, respectively; Table 3). Our event-study estimates are shown in panels C and D of Figure 4. These highlight that CPUE near the high seas pockets is not different from CPUE observed 100–200 nm from the pockets, even after the no-take zones were established. Overall, there is no evidence to indicate that closing the high seas pockets produced spillover benefits to adjacent purse seine vessels targeting skipjack tuna. This null result holds when we estimate the model separately for each pocket (Table C2). Similar results for the longline fleet, other species, and additional model specifications are presented in the Supplementary Materials section B.3.

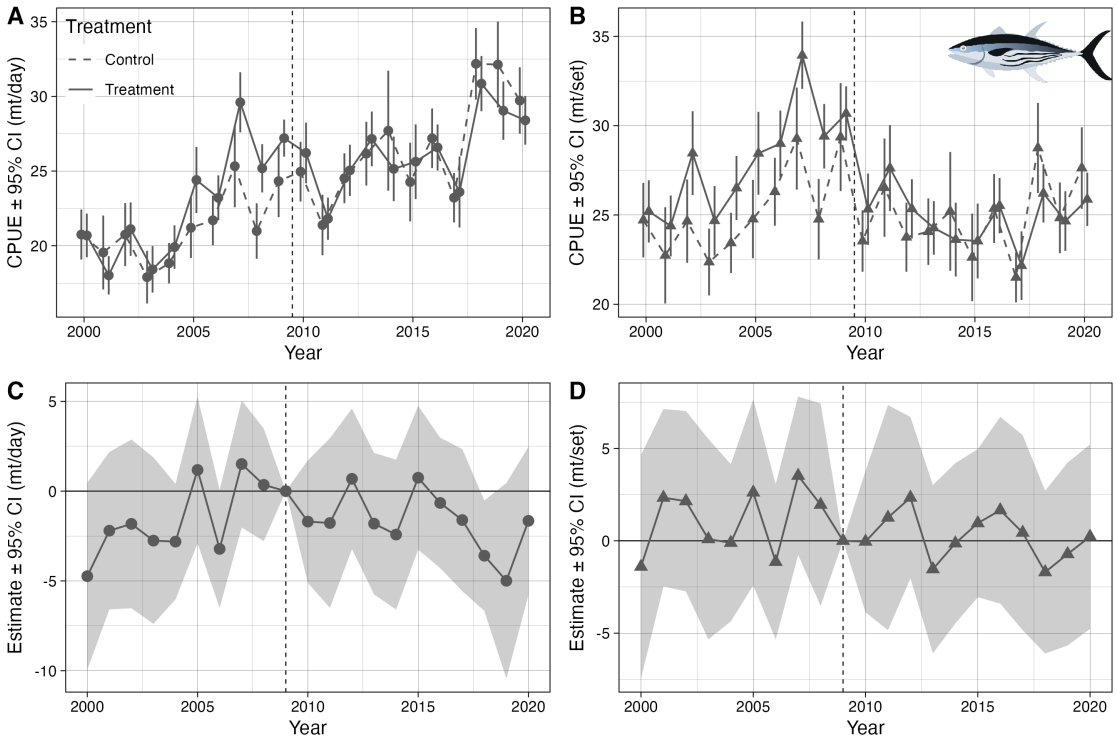



Figure 4: Change in catch-per-unit-effort (CPUE) of skipjack tuna in the purse seine fleet. Panels A and B show time series of mean CPUE in tons per day and tons per set, respectively, across all grid cells within 100 nm of the high seas pockets (solid) and those far away (100–200 nm, dashed). Both the near and far purse seine samples are restricted to grid cells within PNA member Exclusive Economic Zones. Error bars show 95% confidence intervals. Panels C and D show coefficient estimates for the change in skipjack tuna CPUE near the high seas pockets, relative to changes in CPUE observed in areas further away (100–200 nm). Ribbons show 95% confidence intervals.

Table 3: Coefficient estimates for the change in skipjack tuna CPUE in the purse seine fleet operating in areas within 100 nautical miles of the high seas pockets after the closure, relative to the change in CPUE observed for areas between 100 and 200 nautical miles and inside PNA Exclusive Economic Zones.

| | CPUE (mt/day) | CPUE (mt/set) |
|---|---|---|
| Post x Near | −0.496 | −0.865 |
| | (0.680) | (0.818) |
| $\bar{Y}_{pre}$ | 22.946 | 28.182 |
| Num.Obs. | 5960 | 5960 |
| $R^2$ Adj. | 0.333 | 0.245 |

* p < 0.1, ** p < 0.05, *** p < 0.01
The unit of observation is a grid cell in a given year. All model specifications include fixed effects by year and grid cell. Numbers in parentheses are Conley standard errors with a 220km radius. $\bar{Y}_{pre}$ indicates the mean of each outcome variable over the pre-closure period.

## 2.4 Policy Implications for Large-Scale MPAs on the High Seas

We evaluated changes in fishing effort and CPUE inside and near two high seas areas where a closure on industrial purse seine fishing has been in place since 2010. Our results show that purse seine fishing effort was significantly curtailed in the two high seas pockets. This intervention led to longliners catching 19.6% more bigeye tuna per unit effort within the pockets. But, we do not observe evidence that the closure boosted catch rates of skipjack tuna for purse seine vessels fishing near the high seas pockets.

The success in eliminating fishing effort within the high seas pockets highlights the power of the environmental market implemented by the Parties to the Nauru Agreement. Their waters contain some of the most productive skipjack tuna fishing grounds in the world and the majority of the Pacific resource is typically caught in these waters [17, 18]. The combination of high environmental productivity and strong market power gives PNA members considerable bargaining power when negotiating access agreements for foreign fishing vessels. This allowed the PNA members to close a portion of the high seas to purse seine fishing well before the UN High Seas Treaty was even

ratified.

However, the closure also highlights the important role that regional fisheries management organizations play in governing areas beyond national jurisdiction. While PNA members might have initiated the move to close the high seas [22], the closure was supported by the governance scaffolding provided by the WCPFC. For example, the closure was explicitly listed in a Conservation Management Measure issued by the WCPFC [21]. Future efforts to extend MPAs into areas beyond national jurisdiction should consider the multiple roles that regional fisheries management organizations can play, be it for their extensive knowledge, historical data archives, relationships with fishing nations, and ability to enact management measures [40, 41].

Even though the intervention successfully reduced purse seine fishing within the high seas pockets, the evidence of benefits to the longline fleet is somewhat limited. We find that longliners catch more fish, but not more tons of fish, per unit effort. This is broadly aligned with initial expectations. Because purse seiners are no longer incidentally catching juvenile bigeye tuna, more of these fish are available for harvest by the longline fleet. It is plausible that there are cohorts of juvenile bigeye that evade purse seine mortality within the pockets and some survivors later show up on longline hooks. But continued or even higher longline fishing mortality within the pockets would reduce the probability of survival into larger size classes.

The weak and null effects we detect could be due to the resolution of the data. Longline data are reported on a $5^\circ \times 5^\circ$ grid, which may be too coarse to detect the expected small effects [34]. Unfortunately, the WCPFC does not make higher resolution data publicly available but, in principle, these exist and would allow for a more granular test as done elsewhere [42, 5, 6]. Of course, another explanation is that the high seas closures are simply not providing any conservation benefits, suggesting that traditional fisheries management approaches such as regulating the use of drifting Fish Aggregating Devices across the region could have been more effective [34, 16].

We fail to detect any benefit to purse seine fisheries operating near the high seas pockets, even though the closure significantly reduced fishing mortality for skipjack tuna. This may be because longliners continued operating in the region and they occasionally catch skipjack tuna. This lingering source of fishing mortality means that skipjack are only partially protected, and previous work has shown that fully protected MPAs are the ones most likely to deliver conservation

benefits [43].

Differences in habitat are another possible explanation for the null findings compared to MPAs situated inside of EEZs where spillover has been documented [4, 5, 6]. For example, previous work has shown that pelagic species tend to aggregate around shallow (< 150 m deep) seamounts and islands [9, 10, 11, 12, 44]. Most Large-Scale MPAs within EEZs are placed around volcanic islands, so it is not surprising that these MPAs contain multiple islands and shallow seamounts that produce the heterogeneous habitat that aggregates highly mobile pelagic species. The most comprehensive global dataset on seamount location and depth [45, 46] shows that the high seas pockets do not contain any shallow seamounts (Figure A3). This observation lends some support to calls for high seas MPAs to focus on these underwater features [47].

In conclusion, our results show that PNA members were able to use their bargaining power—rooted in successful fisheries management—to significantly reduce fishing effort within the high seas pockets. This intervention led to the world's first *de facto* large-scale MPA on the high seas. We find mixed evidence in terms of the effects of this significant reduction in fishing effort, indicating that benefits from high seas MPAs may be weaker than those observed to date around similar-sized MPAs within EEZs. Our results also highlight how features such as degree of protection, amount of fishing displaced, and presence of critical habitat may drive potential outcomes of future *de jure* MPAs on the high seas.

# 3 Methods

Prior to running any empirical analysis, we pre-registered our analysis plans using the AsPredicted template in the Open Science Framework. Pre-registration plans aim to prevent p-hacking, *ex post* hypothesizing, and discarding important null findings [48, 49]. The pre-registration plan states the hypotheses being tested, the outcome variables and units of measurement, and the process by which conditions (e.g. treated vs. control) are assigned. Our pre-registered plan is available at Villaseñor Derbez and Lynham [35].

We set out to test three hypotheses: (1) the PNA's closure of the high-seas pockets to purse seine fishing reduced purse seine fishing effort within the high seas pockets; (2) reducing purse

seine fishing effort within the high seas pockets will increase catch-per-unit-effort of tuna, especially bigeye tuna, for the longline fleet operating within the high seas pockets; and (3) the closure will also increase catch-per-unit-effort of bigeye, skipjack, and yellowfin tuna for the purse seine and longline fleets operating in PNA waters within 100 nautical miles of the high seas pockets, relative to catch-per-unit-effort 100-200 nautical miles from the high seas pockets. The methods below describe the steps we take to build our panel dataset and test our hypotheses.

## 3.1 Data sources

We used publicly available data provided by the Western and Central Pacific Fisheries Commission (WCPFC). We downloaded data from the WCPFC Public Domain data portal.[4] Specifically, the purse seine data came from the following file: *"Aggregated data, grouped by 1˚x1˚ latitude/longitude grids, YEAR and MONTH"* (WCPFC_S_PUBLIC_BY_1x1_MM5). And the longline data came from *"Aggregated data, grouped by 5˚x5˚ latitude/longitude grids, YEAR and MONTH"* (WCPFC_L_PUBLIC_BY_YR_MON_3). Both datasets were downloaded on Oct 31, 2025.

## 3.2 Dataset construction

### 3.2.1 Outcomes of interest

Our two outcomes of interest are fishing effort and catch-per-unit-effort (CPUE). We measured fishing effort by the purse seine fleet in both days and sets. These are reported monthly as 1° by 1° grid cell totals. We summed the monthly totals to obtain annual measures of effort by grid cell.

We calculate CPUE per grid cell and year as the sum of the total catch divided by the total effort reported for that cell, adjusting for gear-specific differences. For purse seine vessels, CPUE is measured in tons of fish per day and tons of fish per set. Longline CPUE is measured in number of fish per thousand hooks and tons per thousand hooks.

[4] `www.wcpfc.int/wcpfc-public-domain-aggregated-catcheffort-data-download-page`

## 3.3 Treatment and Control Groups

All regression-based hypothesis tests include indicator variables for the periods before and after the high seas closures (effective January 2010), as well as for treatment and control groups. The definition of treatment and control groups varies by hypothesis test. When testing for changes in purse seine fishing effort, we define treated grid cells as those within the high seas pockets, and control grid cells as other high seas grid cells that fall under the WCPFC convention area and lie between 20°N and 20°S. When testing for changes in CPUE to the longline fleet operating in the high seas pockets, we consider grid cells inside high seas pockets as "treated" and other high seas grid cells that fall under the WCPFC convention area and lie between 20°N and 20°S as "control". Finally, when testing for spillover benefits, we follow the same Before-After-Near-Far design used to test for spillover benefits from Large-Scale Marine Protected Areas in other settings [4, 5]. Specifically, grid cells within 100 nautical miles of the high seas pockets' boundaries are labeled as treated. Grid cells between 100 and 200 nautical miles from the boundaries are labeled as controls. For the purse seine fishery, we only retain grid cells that fall within PNA countries' EEZs to account for the sharp difference in management between vessels that operate under the vessel-day scheme *vs.* those fishing on the high seas [17, 18, 19].

## 3.4 Identification and Estimation

### 3.4.1 Testing for changes in fishing effort inside the high seas pockets

Let $E_{it}$ be purse seine effort (measured in sets or days) in grid cell $i$ at time $t$. We first test for changes in fishing effort in the high seas pockets with a simple linear regression that does not account for changes in region-wide fishing effort. The linear regression has the following form:

$$E_{it} = \sum_{t \neq 0} \beta_t \mathrm{T}_t + \mu_i + \epsilon_{it} \tag{1}$$

where $\mathrm{T}_t$ are indicators for periods relative to year of treatment, such that t = 0 corresponds to the last year when purse seine fishing was allowed, 2009. Negative values of t are pre-closure years, and positive values of t are post-closure years. The parameter of interest is $\beta_t$, which is a vector of

coefficients capturing changes in effort inside the high seas pockets relative to the last year when fishing was allowed (2009). Then, $\mu_i$ are grid-cell fixed-effects and $\epsilon_{it}$ is the error term, which accounts for spatio-temporal autocorrelation via Conley standard errors. These results are shown in Figure B1, which captures changes in effort relative to 2009. However, our main specification estimates changes across time, relative to changes in effort in other high seas grid cells:

$$E_{it} = \sum_{t \neq 0} \beta_t \left( \mathrm{T}_t \times \mathrm{D}_i \right) + \mu_i + \tau_t + \epsilon_{it}. \tag{2}$$

Here, $D_i = 1$ indicates grid cell $i$ falls within the high seas closure and $D_i = 0$ indicates other high seas cells not subject to the closure. $\tau_t$ are fixed-effects by year and $\mu_i$ is defined as above. This model allows us to account for region-wide trends in fishing effort, and under standard assumptions, allows us to derive a causal estimate of the intervention. The results are shown in Figure 2 Panels C and D.

Aggregate treatment effects were also estimated using the following regression models: $E_{it} = \beta \mathrm{Post}_t + \mu_i + \epsilon_{it}$ and $E_{it} = \beta \mathrm{Post}_t \times \mathrm{D}_i + \mu_i + \tau_t + \epsilon_{it}$. $\mathrm{Post}_t$ is a dummy variable that takes the value of 1 or 0 if the observation comes from after or before the implementation period, respectively. Results are shown in Tables 1, B1, and B2.

As noted in our preregistration plan, we also estimate models using inverse hyperbolic sine -transformed outcomes. The transformation is similar to a log-transformation, but accommodates outcome values at 0 and is considered a standard method for cases like this [50]. Event-study plots are shown in Figures B1 and B2, while aggregate results are shown in Tables B1 and B2.

### 3.4.2 Testing for changes in CPUE inside the high seas pockets

Let $y_{it}$ denote longline catch-per-unit-effort (CPUE) in grid cell $i$ at time $t$. We estimate:

$$y_{it} = \sum_{t \neq 0} \beta_t \left( \mathrm{T}_t \times \mathrm{D}_i \right) + \mu_i + \tau_t + \epsilon_{it} \tag{3}$$

where all regressors have the same definitions as above. The coefficient of interest, $\beta_t$, captures the change in CPUE within the high seas pocket relative to changes observed in other high seas grid cells where purse seine effort was not curtailed. A pre-post model with the form $y_{it} = \beta \mathrm{Post}_t \times$

$\mathrm{D}_i + \mu_i + \tau_t + \epsilon_{it}$ was estimated to test for aggregate effects. We also estimated log-linear versions of this model. Results for bigeye tuna are shown in Figure 3 Panels C and D and in Table 2. Results for other specifications and other species are shown in the Supplementary Materials section B.2 in Tables B3–B6 and Figures B3–B6.

### 3.4.3 Testing for spillover benefits

Let $y_{it}$ denote longline or purse seine catch-per-unit-effort (CPUE) in grid cell $i$ at time $t$. We estimate the following regression:

$$y_{it} = \sum_{t \neq 0} \beta_t \left( \mathrm{T}_t \times \mathrm{D}_i \right) + \mu_i + \tau_t + \epsilon_{it} \tag{4}$$

where the coefficient of interest, $\beta_t$, captures the dynamic treatment effect of the high seas closure on CPUE within 100 nautical miles of the high seas pockets, relative to changes observed in other grid cells between 100 and 200 nautical miles from the high seas pockets. A pre-post model with the form $y_{it} = \beta \mathrm{Post}_t \times \mathrm{D}_i + \mu_i + \tau_t + \epsilon_{it}$ was also estimated to test for aggregate effects along with log-linear specifications. Results for skipjack tuna in the purse seine fleet are shown in Figure 4 Panels C and D and in Table 3. Results for other specifications and species are shown in Supplementary Materials section B.3 in Tables B7–B14 and Figures B7–B14.

### 3.4.4 Weights

By construction, $y_{it} = \frac{\sum_{j \in (i,t)} \mathrm{Catch}_j}{\sum_{j \in (i,t)} \mathrm{Effort}_j}$, so $\mathrm{Var}(y_{it}) \propto \frac{1}{\mathrm{Effort}_{it}}$. We therefore assume

$$\mathrm{Var}(\epsilon_{it} \mid X) = \frac{\sigma^2}{\omega_{it}}, \qquad \omega_{it} = \mathrm{Effort}_{it},$$

and estimate all CPUE models via weighted least squares using effort as weights. The measure of effort used as weight will be the same as the measure of effort used in the denominator in the CPUE calculation, which varies by gear type as stated in section 3: Dependent variable.

### 3.4.5 Software and Reproducibility

All data cleaning and sample construction was performed with the tidyverse (v2.0.0), sf (1.1-2), and terra (1.9-34) packages using R (v4.6.1) and RStudio (2026.08.1 Build 195 "Yellow Yarrow" [51, 52, 53, 54, 55]). All models were fit using the fixest package (v 0.14.2; [56]). Regression tables were built using the modelsummary package (v2.6.0; [57]), and event-study plots were built with ggfixest (v0.4.0, [58]) and customized with the ggplot2 package (v4.0.3, [59]). All our data and code are publicly available in a GitHub repository posted at `https://github.com/human-ocean/PNA_HS_closure`.

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# Appendix

Supplementary Materials

For

**Limited Conservation Benefits From the First Marine Protected Area in the High Seas**

# A Supplementary Figures and Tables

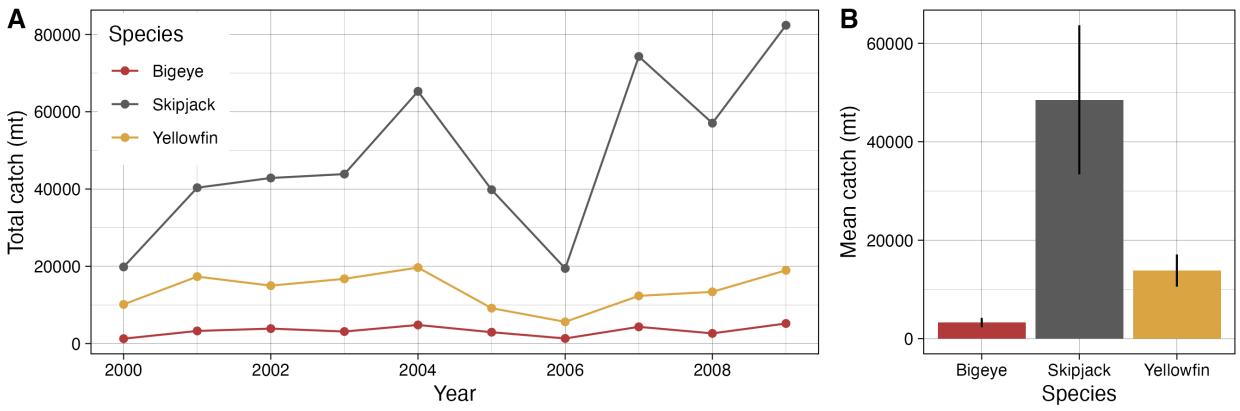



Figure A1: Historical (2000-2009) catch of bigeye, skipjack, and yellowfin tuna by tuna purse seiners fishing in the high seas pockets before their closure. Panel A shows a time series of catch by species, panel B shows mean (± SD) catch by species.

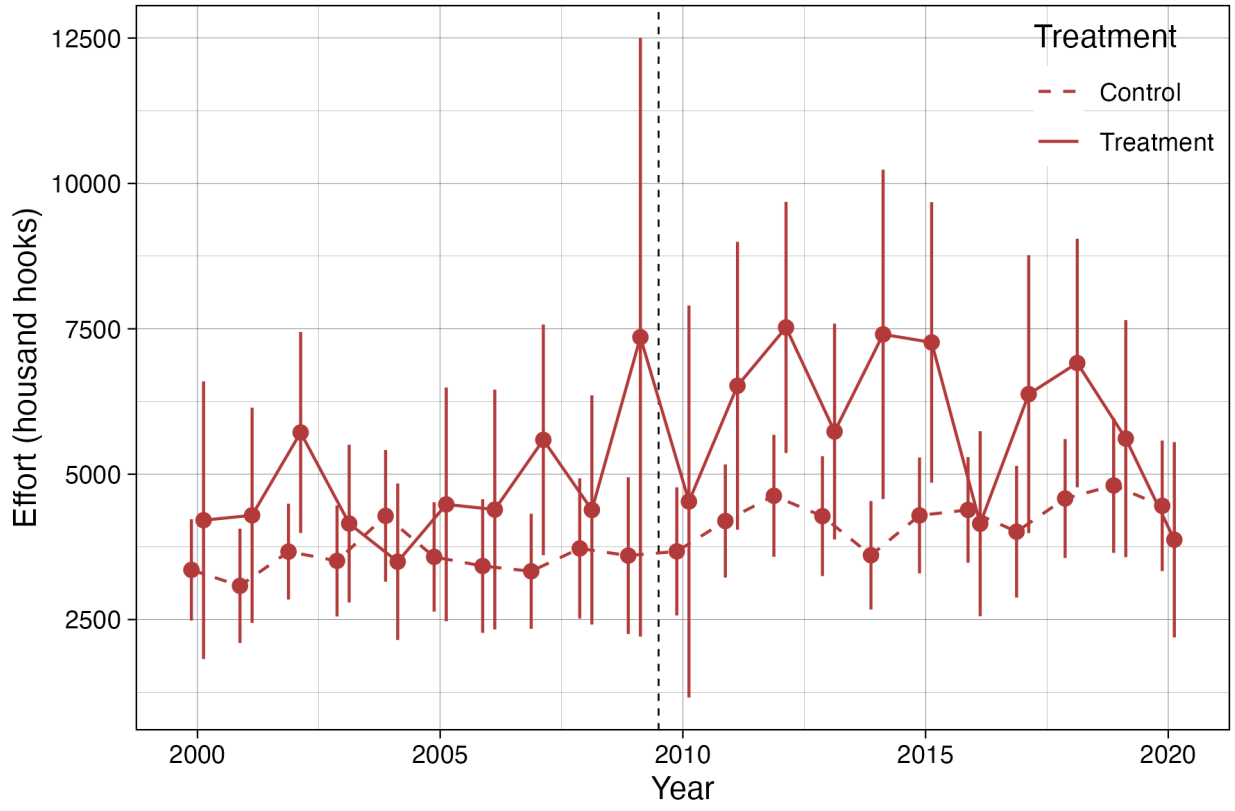



Figure A2: Time series of mean fishing effort (thousand hooks) by the longline fleet operating in the high seas pockets (solid line) and other high seas grid cells (dashed line).

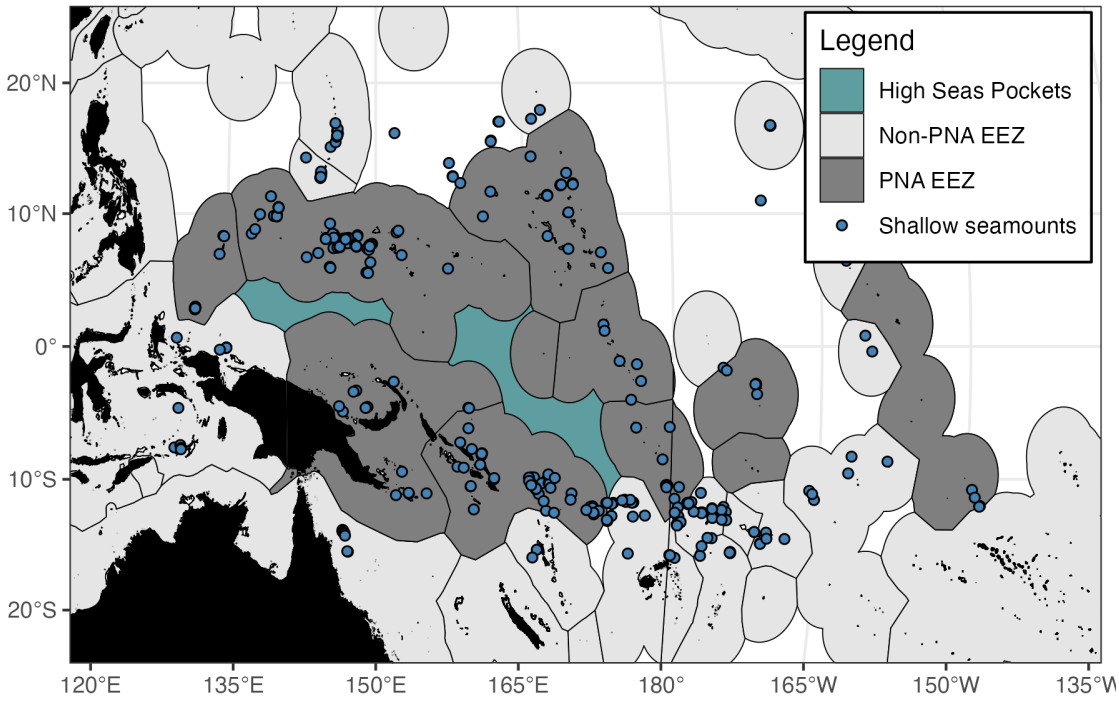



Figure A3: Map of shallow seamounts in the Western and Central Pacific. Blue points indicate the centroid of shallow seamounts ($\leq$ 150 m deep) as defined in [45, 46]. Light gray polygons show the Exclusive Economic Zones of all nations. Dark gray polygons highlight the Exclusive Economic Zones of PNA nations and Tokelau. The two teal polygons show the High Seas Pockets closed to purse seine fishing starting in 2010.

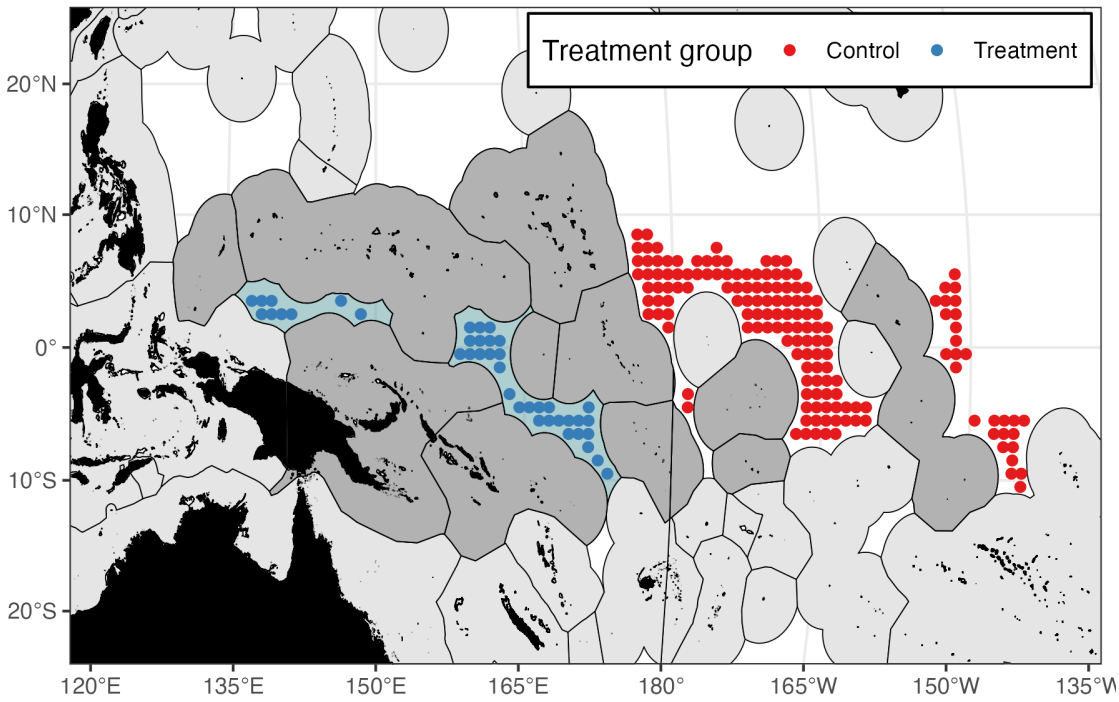



Figure A4: Map of location of treatment and control cells used in our test of changes in purse seine fishing effort in the high seas pockets.

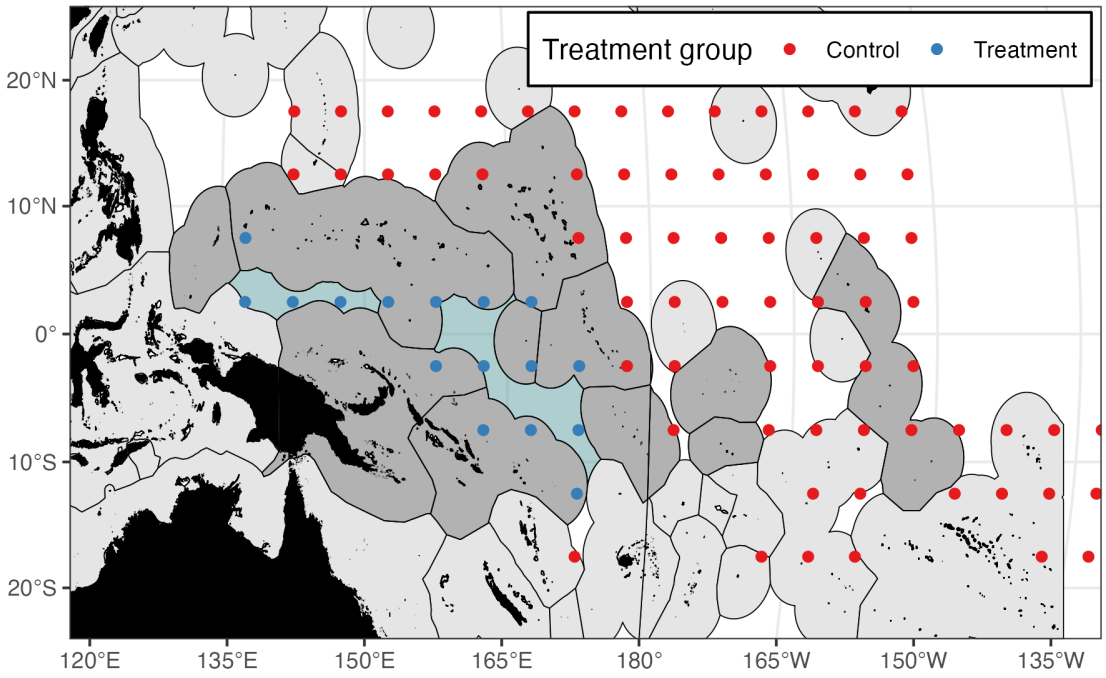



Figure A5: Map of location of treatment and control grid cells used in our test of changes in CPUE for the longline fleet operating in the high seas pockets.

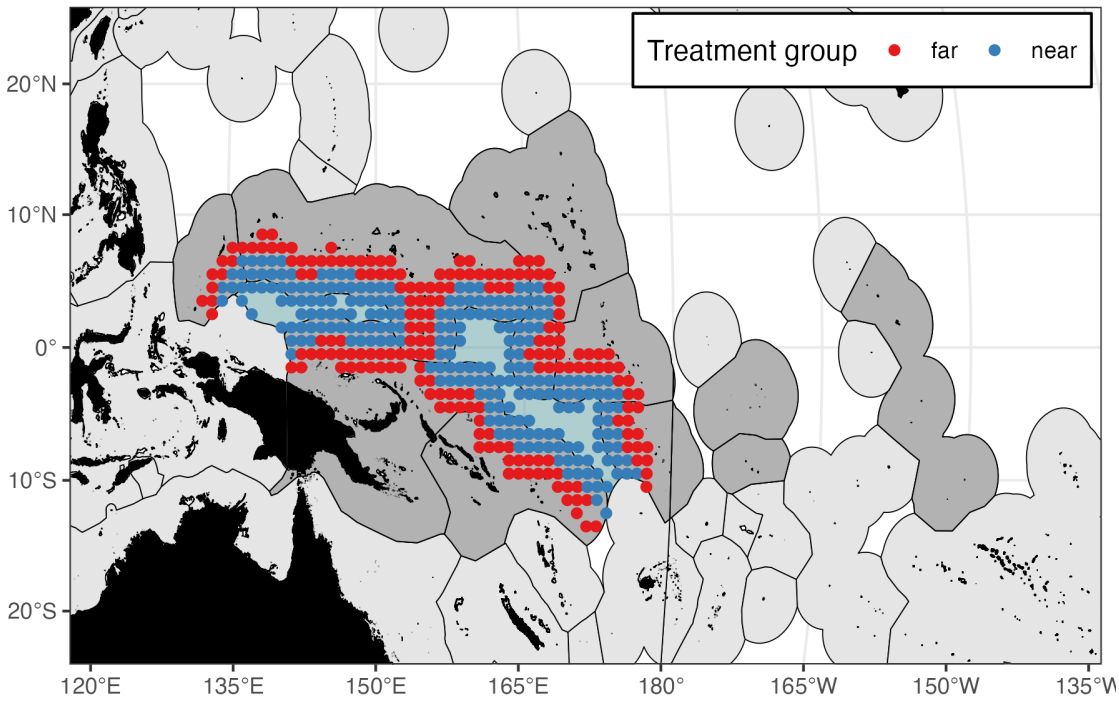



Figure A6: Map of location of treatment and control grid cells used in our test of a spillover benefit to the purse seine fleet.

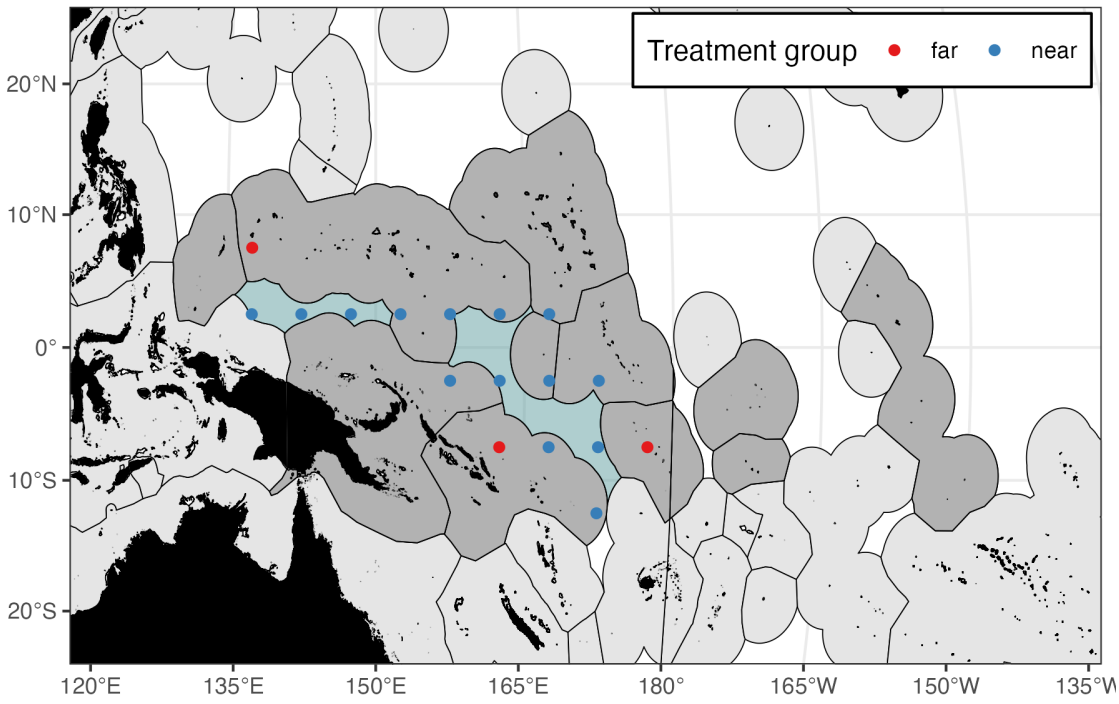



Figure A7: Map of location of treatment and control grid cells used in our test of a spillover benefit to the longline fleet.

# B Additional Results

## B.1 Decrease in purse seine effort in high seas pockets

The reduction in purse seine fishing effort in the high seas pockets is robust across model specifications and outcome transformations. A within-treated specification, which estimates changes relative to the 2009 baseline without a control comparison, shows significant reductions of 54.55 days and 44.02 sets per grid cell per year ($p < 0.01$; Table B1), with event-study estimates confirming no pre-treatment trends (Figure B1). Results comparing trends with other high seas areas within the WCPFC convention area also remain significant when outcomes are transformed using the inverse hyperbolic sine (Table B2; Figure B2).

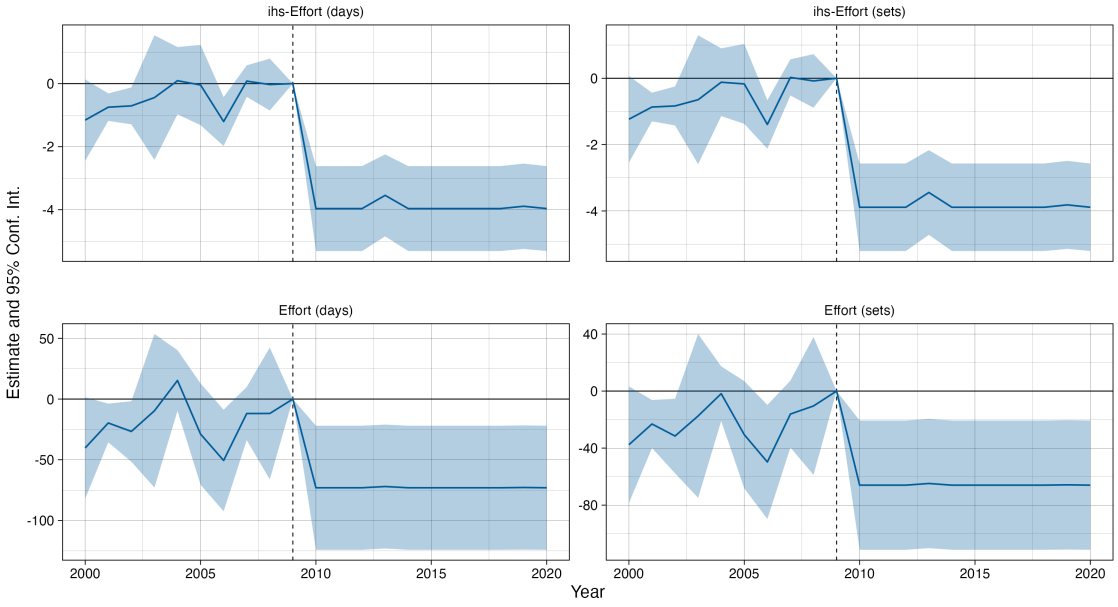



Figure B1: Event-study estimates of the change in purse seine fishing effort within the high seas pockets, estimated using only treated grid cells relative to the 2009 baseline (without a control comparison). Estimates are shown for fishing effort measured in days (left panels) and number of sets (right panels), under the inverse hyperbolic sine (IHS) transformation (top panels) and in levels (bottom panels).

Table B1: Coefficient estimates for the change in fishing effort inside the high seas pockets after the closure. Panel A presents results in levels. Panel B presents results in which the dependent variable is transformed using the inverse hyperbolic sine (IHS) transformation.

| | Effort (days) | Effort (sets) |
|---|---|---|
| *A) Levels* | | |
| Post | -54.555*** | -44.018*** |
| | (13.647) | (10.400) |
| Num.Obs. | 840 | 840 |
| $R^2$ Adj. | 0.363 | 0.332 |
| *B) Inverse-hyperbolic sine transformation* | | |
| Post | -3.505*** | -3.310*** |
| | (0.452) | (0.424) |
| Num.Obs. | 840 | 840 |
| $R^2$ Adj. | 0.675 | 0.662 |

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$
The unit of observation is a grid cell in a given year. Numbers in parentheses are Conley standard errors using a 220km radius.

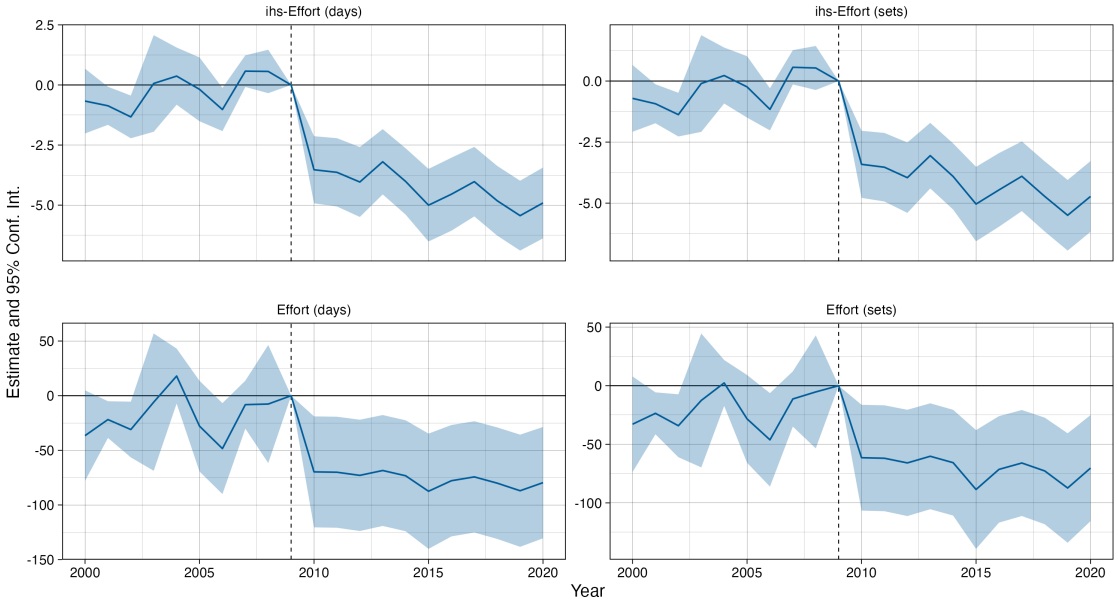



Figure B2: Event-study estimates of the change in purse seine fishing effort in the high seas pockets, relative to fishing effort in comparable high seas regions in the WCPFC convention area. Estimates are shown for fishing effort measured in days (left panels) and number of sets (right panels), under the inverse hyperbolic sine (IHS) transformation (top panels) and in levels (bottom panels). The bottom panels are equivalent to the main text results shown in Figure 2.

Table B2: Coefficient estimates for the change in fishing effort inside the high seas pockets after the closure, relative to the change in fishing effort observed in other comparable high seas areas of the WCPFC convention area. Panel A presents results in levels (identical to the main-text estimates in Table 1). Panel B presents results in which the dependent variable is transformed using the inverse hyperbolic sine (IHS) transformation.

| | Effort (days) | Effort (sets) |
|---|---|---|
| *A) Levels* | | |
| Post x Treated | -59.470*** | -50.933*** |
| | (13.763) | (10.608) |
| Num.Obs. | 4011 | 4011 |
| $R^2$ Adj. | 0.388 | 0.301 |
| *B) Inverse-hyperbolic sine transformation* | | |
| Post x Treated | -4.036*** | -3.881*** |
| | (0.478) | (0.451) |
| Num.Obs. | 4011 | 4011 |
| $R^2$ Adj. | 0.465 | 0.451 |

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

The unit of observation is a grid cell in a given year. All model specifications include fixed effects by year and grid cell. Numbers in parentheses are Conley standard errors using a 220km radius.

## B.2 Change in longline CPUE in high seas pockets

Here we present results for changes in longline CPUE within the high seas pockets for all species and additional model specifications. Across all species combined (Table B3; Figure B3) and for albacore tuna (Table B4; Figure B4), we detect no significant change in longline CPUE. The bigeye tuna results shown in the main text are qualitatively similar under log-transformation, though we note a marginally significant decline in CPUE when measured in metric tons (Table B5; Figure B5). For yellowfin tuna, we estimate a significant increase in log-transformed CPUE measured in number of fish ($p < 0.05$; Table B6; Figure B6).

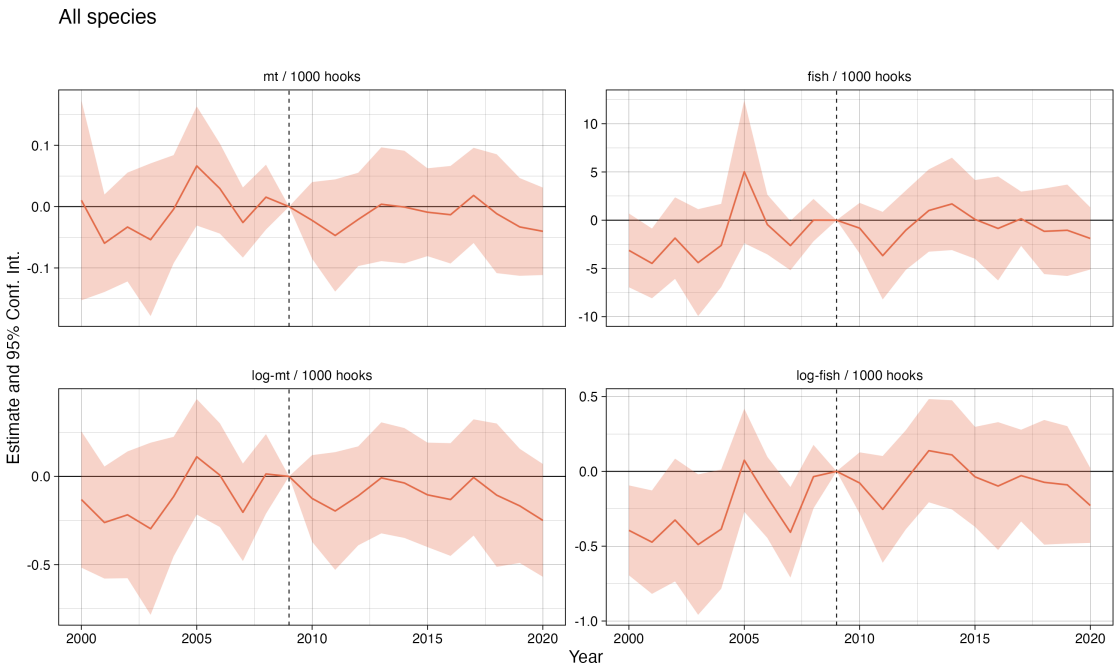



Figure B3: Event-study estimates of the change in longline catch-per-unit-effort (CPUE) across all species combined in the high seas pockets, relative to CPUE in other tropical (20°S to 20°N) high seas regions in the WCPFC convention area. Estimates are shown for CPUE measured in metric tons per 1000 hooks (left panels) and number of fish per 1000 hooks (right panels), in levels (top panels) and for log-transformed CPUE (bottom panels).

Table B3: Coefficient estimates for the change in all tuna CPUE in the longline fleet in the high seas pockets after the closure, relative to the change in CPUE observed for other tropical (20°S - 20°N) high seas areas in the WCPFC convention area. Panel A presents results in levels. Panel B presents results in which the dependent variable is log-transformed.

| | fish / 1000 hooks | mt / 1000 hooks |
|---|---|---|
| *A) Levels* | | |
| Post x Treated | 0.664 | -0.009 |
| | (1.605) | (0.028) |
| Num.Obs. | 1632 | 1632 |
| $R^2$ Adj. | 0.637 | 0.507 |
| *B) Log-transformed* | | |
| Post x Treated | 0.187 | -0.003 |
| | (0.139) | (0.089) |
| Num.Obs. | 1632 | 1632 |
| $R^2$ Adj. | 0.700 | 0.616 |

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

The unit of observation is a grid cell in a given year. All model specifications include fixed effects by year and grid cell. Numbers in parentheses are Conley standard errors with a 1100km radius.

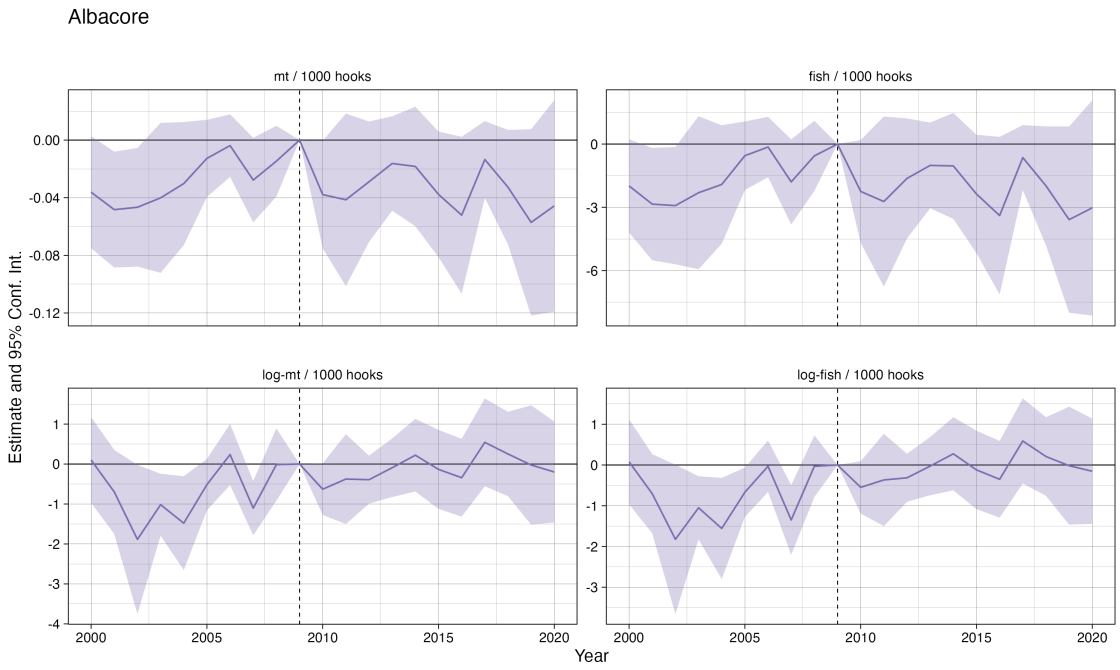



Figure B4: Event-study estimates of the change in longline catch-per-unit-effort (CPUE) for albacore tuna in the high seas pockets, relative to CPUE in other tropical (20°S to 20°N) high seas regions in the WCPFC convention area. Estimates are shown for CPUE measured in metric tons per 1000 hooks (left panels) and number of fish per 1000 hooks (right panels), in levels (top panels) and for log-transformed CPUE (bottom panels).

Table B4: Coefficient estimates for the change in albacore tuna CPUE in the longline fleet in the high seas pockets after the closure, relative to the change in CPUE observed for other tropical (20°S - 20°N) high seas areas in the WCPFC convention area. Panel A presents results in levels. Panel B presents results in which the dependent variable is log-transformed.

| | fish / 1000 hooks | mt / 1000 hooks |
|---|---|---|
| *A) Levels* | | |
| Post x Treated | -0.645 | -0.009 |
| | (0.793) | (0.012) |
| Num.Obs. | 1606 | 1606 |
| $R^2$ Adj. | 0.856 | 0.852 |
| *B) Log-transformed* | | |
| Post x Treated | 0.644 | 0.540 |
| | (0.499) | (0.485) |
| Num.Obs. | 1606 | 1603 |
| $R^2$ Adj. | 0.821 | 0.809 |

* p < 0.1, ** p < 0.05, *** p < 0.01

The unit of observation is a grid cell in a given year. All model specifications include fixed effects by year and grid cell. Numbers in parentheses are Conley standard errors with a 1100km radius.

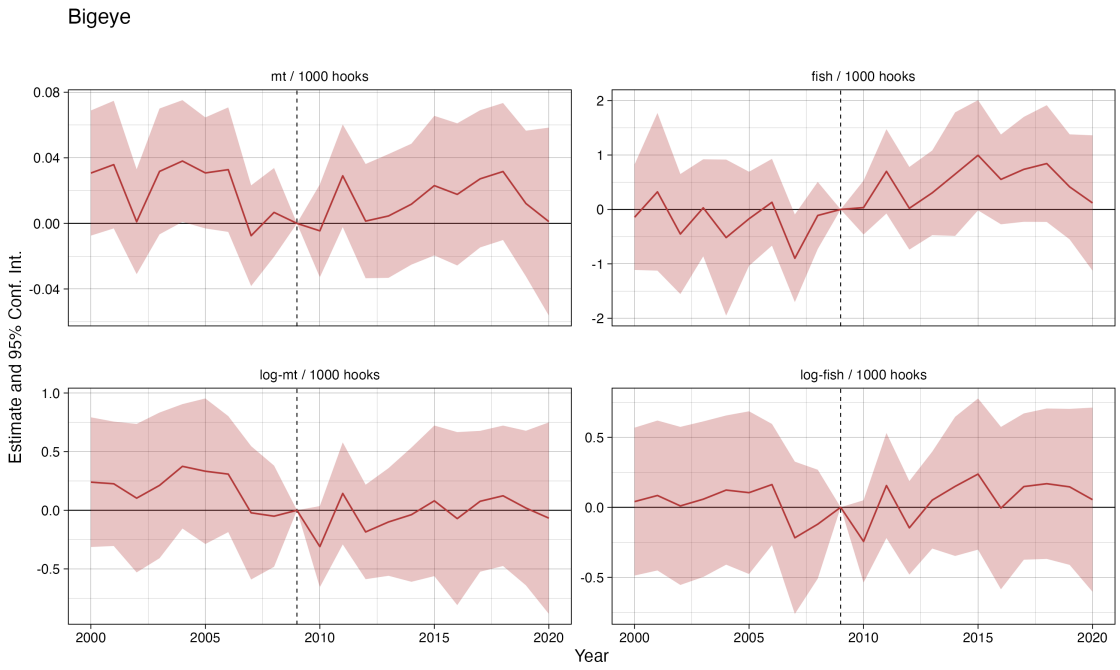



Figure B5: Event-study estimates of the change in longline catch-per-unit-effort (CPUE) for bigeye tuna in the high seas pockets, relative to CPUE in other tropical (20°S to 20°N) high seas regions in the WCPFC convention area. Estimates are shown for CPUE measured in metric tons per 1000 hooks (left panels) and number of fish per 1000 hooks (right panels), in levels (top panels) and for log-transformed CPUE (bottom panels). The top panels are equivalent to the main text results shown in Figure 3.

Table B5: Coefficient estimates for the change in bigeye tuna CPUE in the longline fleet in the high seas pockets after the closure, relative to the change in CPUE observed for other tropical (20°S - 20°N) high seas areas in the WCPFC convention area. Panel A presents results in levels (same as in Table 2). Panel B presents results in which the dependent variable is log-transformed.

| | fish / 1000 hooks | mt / 1000 hooks |
|---|---|---|
| *A) Levels* | | |
| Post x Treated | 0.697* | -0.003 |
| | (0.363) | (0.011) |
| Num.Obs. | 1631 | 1631 |
| $R^2$ Adj. | 0.798 | 0.829 |
| *B) Log-transformed* | | |
| Post x Treated | 0.055 | -0.176* |
| | (0.106) | (0.094) |
| Num.Obs. | 1631 | 1631 |
| $R^2$ Adj. | 0.852 | 0.880 |

* p < 0.1, ** p < 0.05, *** p < 0.01

The unit of observation is a grid cell in a given year. All model specifications include fixed effects by year and grid cell. Numbers in parentheses are Conley standard errors with a 1100km radius.

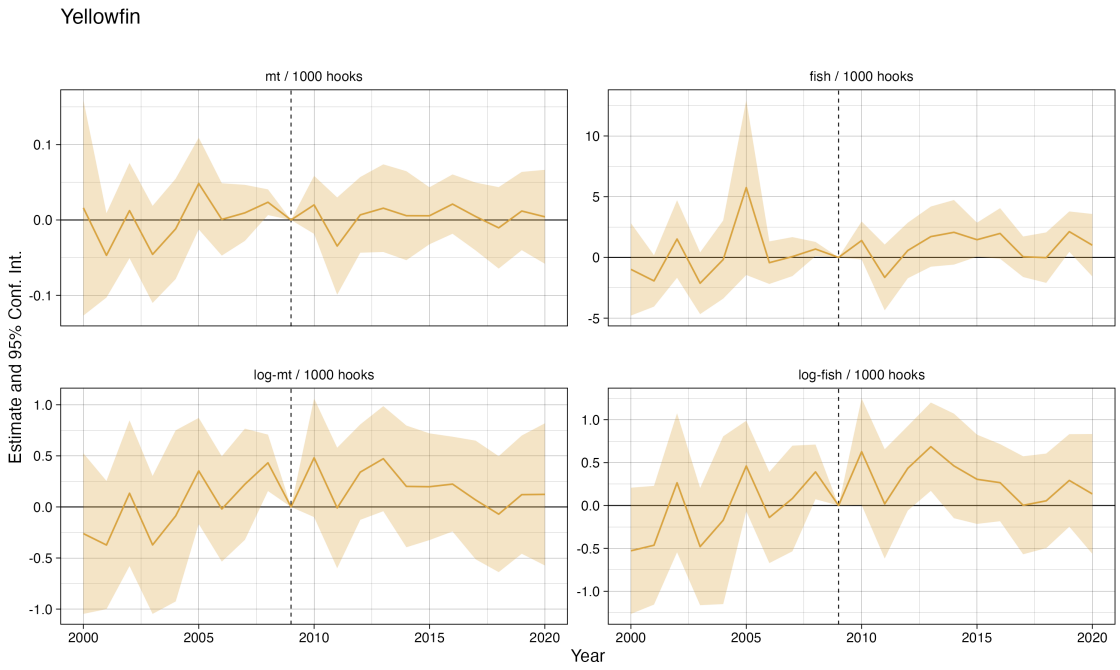



Figure B6: Event-study estimates of the change in longline catch-per-unit-effort (CPUE) for yellowfin tuna in the high seas pockets, relative to CPUE in other tropical (20°S to 20°N) high seas regions in the WCPFC convention area. Estimates are shown for CPUE measured in metric tons per 1000 hooks (left panels) and number of fish per 1000 hooks (right panels), in levels (top panels) and for log-transformed CPUE (bottom panels).

Table B6: Coefficient estimates for the change in yellowfin tuna CPUE in the longline fleet in the high seas pockets after the closure, relative to the change in CPUE observed for other tropical (20°S - 20°N) high seas areas in the WCPFC convention area. Panel A presents results in levels. Panel B presents results in which the dependent variable is log-transformed.

| | fish / 1000 hooks | mt / 1000 hooks |
|---|---|---|
| *A) Levels* | | |
| Post x Treated | 0.598 | 0.002 |
| | (0.857) | (0.016) |
| Num.Obs. | 1632 | 1632 |
| $R^2$ Adj. | 0.538 | 0.607 |
| *B) Log-transformed* | | |
| Post x Treated | 0.329** | 0.171 |
| | (0.151) | (0.118) |
| Num.Obs. | 1632 | 1632 |
| $R^2$ Adj. | 0.702 | 0.704 |

* p < 0.1, ** p < 0.05, *** p < 0.01

The unit of observation is a grid cell in a given year. All model specifications include fixed effects by year and grid cell. Numbers in parentheses are Conley standard errors with a 1100km radius.

## B.3 Spillover benefits to fisheries adjacent to high seas pockets

**Purse seine.** We find no evidence of spillover benefits to the purse seine fleet operating near the high seas pockets for any species or model specification. Results are null across all species combined (Table B7; Figure B7), bigeye tuna (Table B8; Figure B8), skipjack tuna (Table B9; Figure B9), and yellowfin tuna (Table B10; Figure B10), in both levels and log-transformed specifications.

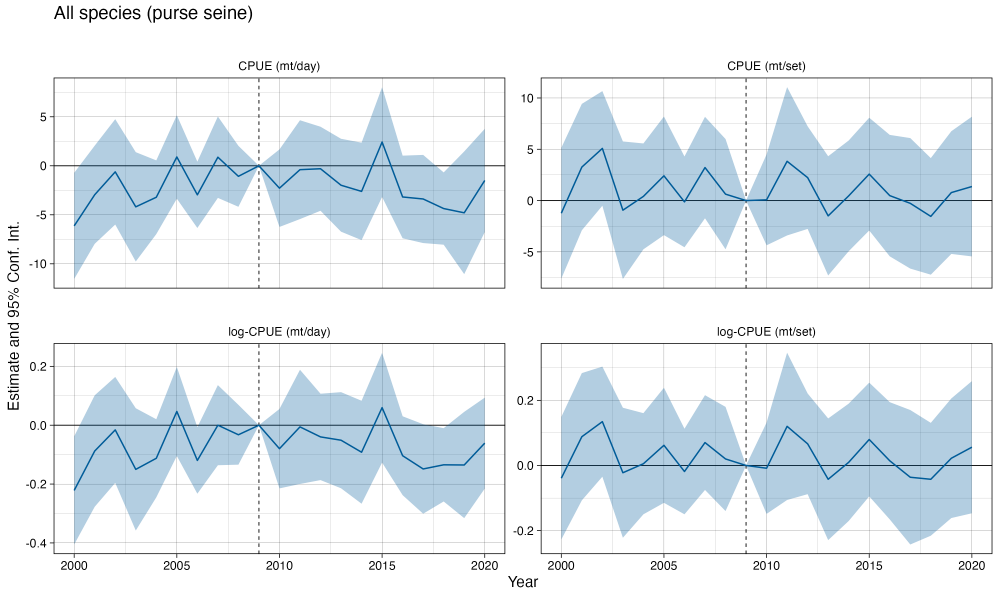



Figure B7: Event-study estimates of the change in purse seine catch-per-unit-effort (CPUE) across all species combined within 100 nm of the high seas pockets, relative to changes in CPUE observed in areas further away (100–200 nm). Both the near (treated) and far (control) samples are restricted to grid cells within PNA member Exclusive Economic Zones. Estimates are shown for CPUE measured in tons per day (left panels) and tons per set (right panels), in levels (top panels) and for log-transformed CPUE (bottom panels).

Table B7: Coefficient estimates for the change in all tuna CPUE in the purse seine fleet in areas within 100 nautical miles of the high seas pockets after the closure, relative to the change in CPUE observed for areas between 100 and 200 nautical miles and inside PNA Exclusive Economic Zones. Panel A presents results in levels. Panel B presents results in which the dependent variable is log-transformed.

| | (1) | (2) |
|---|---|---|
| *A) Levels* | | |
| Post x Near | -0.294 | -0.468 |
| | (0.827) | (1.050) |
| Num.Obs. | 5960 | 5960 |
| $R^2$ Adj. | 0.273 | 0.209 |
| *B) Log-transformed* | | |
| Post x Near | -0.007 | -0.007 |
| | (0.028) | (0.034) |
| Num.Obs. | 5960 | 5960 |
| $R^2$ Adj. | 0.258 | 0.221 |

* p < 0.1, ** p < 0.05, *** p < 0.01

The unit of observation is a grid cell in a given year. All model specifications include fixed effects by year and grid cell. Numbers in parentheses are Conley standard errors with a 220km radius.

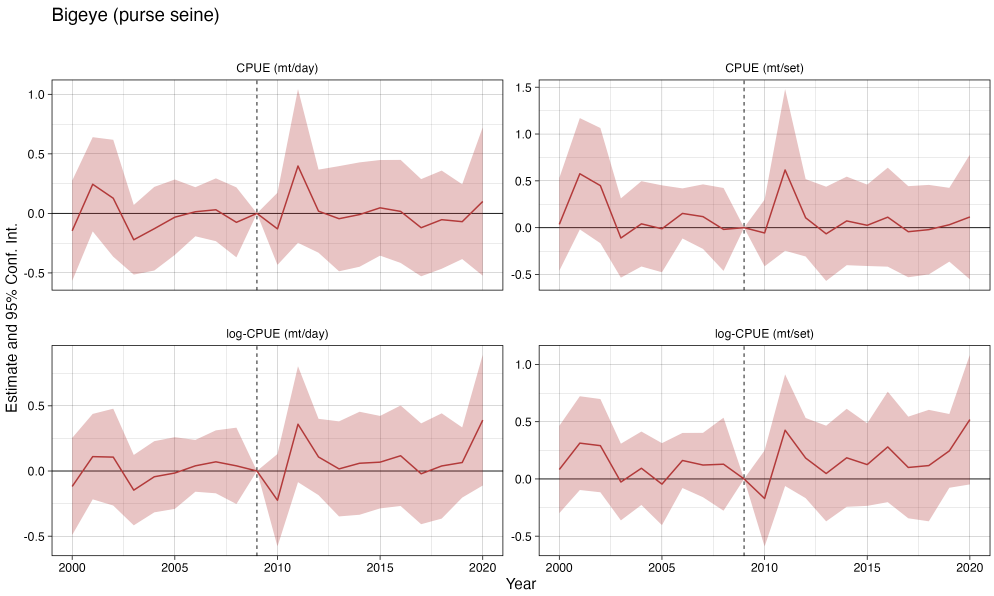



Figure B8: Event-study estimates of the change in purse seine catch-per-unit-effort (CPUE) for bigeye tuna within 100 nm of the high seas pockets, relative to changes in CPUE observed in areas further away (100–200 nm). Both the near (treated) and far (control) samples are restricted to grid cells within PNA member Exclusive Economic Zones. Estimates are shown for CPUE measured in tons per day (left panels) and tons per set (right panels), in levels (top panels) and for log-transformed CPUE (bottom panels).

Table B8: Coefficient estimates for the change in bigeye tuna CPUE in the purse seine fleet in areas within 100 nautical miles of the high seas pockets after the closure, relative to the change in CPUE observed for areas between 100 and 200 nautical miles and inside PNA Exclusive Economic Zones. Panel A presents results in levels. Panel B presents results in which the dependent variable is log-transformed.

| | CPUE (mt/day) | CPUE (mt/set) |
|---|---|---|
| *A) Levels* | | |
| Post x Near | 0.055 | -0.005 |
| | (0.104) | (0.121) |
| Num.Obs. | 5908 | 5908 |
| $R^2$ Adj. | 0.174 | 0.209 |
| *B) Log-transformed* | | |
| Post x Near | 0.086 | 0.081 |
| | (0.086) | (0.096) |
| Num.Obs. | 5908 | 5908 |
| $R^2$ Adj. | 0.227 | 0.279 |

* p < 0.1, ** p < 0.05, *** p < 0.01

The unit of observation is a grid cell in a given year. All model specifications include fixed effects by year and grid cell. Numbers in parentheses are Conley standard errors with a 220km radius.

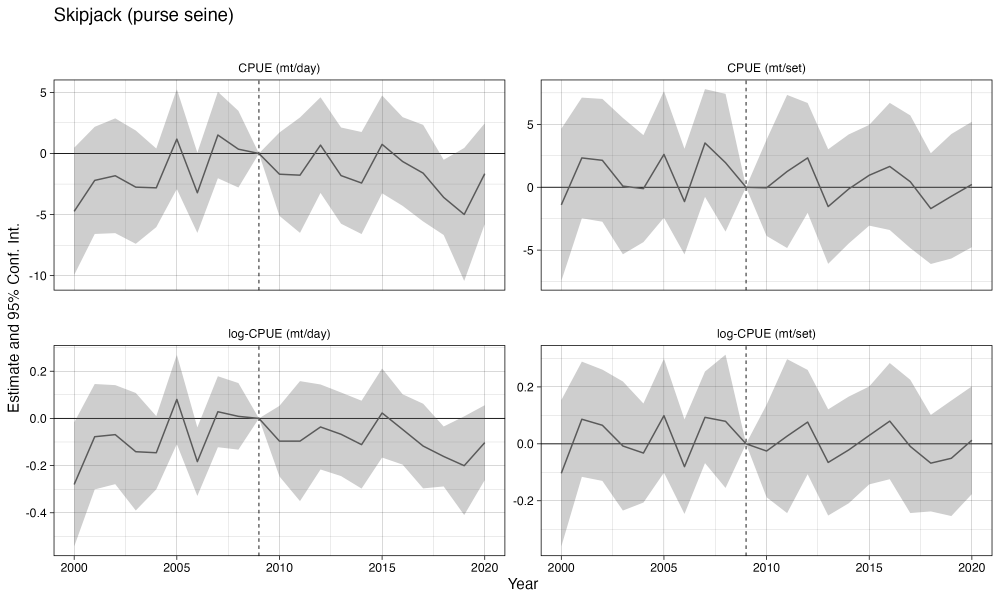



Figure B9: Event-study estimates of the change in purse seine catch-per-unit-effort (CPUE) for skipjack tuna within 100 nm of the high seas pockets, relative to changes in CPUE observed in areas further away (100–200 nm). Both the near (treated) and far (control) samples are restricted to grid cells within PNA member Exclusive Economic Zones. Estimates are shown for CPUE measured in tons per day (left panels) and tons per set (right panels), in levels (top panels) and for log-transformed CPUE (bottom panels). The top panels are equivalent to the main text results shown in Figure 4.

Table B9: Coefficient estimates for the change in skipjack tuna CPUE in the purse seine fleet in areas within 100 nautical miles of the high seas pockets after the closure, relative to the change in CPUE observed for areas between 100 and 200 nautical miles and inside PNA Exclusive Economic Zones. Panel A presents results in levels. Panel B presents results in which the dependent variable is log-transformed.

| | CPUE (mt/day) | CPUE (mt/set) |
|---|---|---|
| *A) Levels* | | |
| Post x Near | -0.496 | -0.865 |
| | (0.680) | (0.818) |
| Num.Obs. | 5960 | 5960 |
| $R^2$ Adj. | 0.333 | 0.245 |
| *B) Log-transformed* | | |
| Post x Near | -0.024 | -0.026 |
| | (0.031) | (0.038) |
| Num.Obs. | 5960 | 5960 |
| $R^2$ Adj. | 0.316 | 0.244 |

* p < 0.1, ** p < 0.05, *** p < 0.01

The unit of observation is a grid cell in a given year. All model specifications include fixed effects by year and grid cell. Numbers in parentheses are Conley standard errors with a 220km radius.

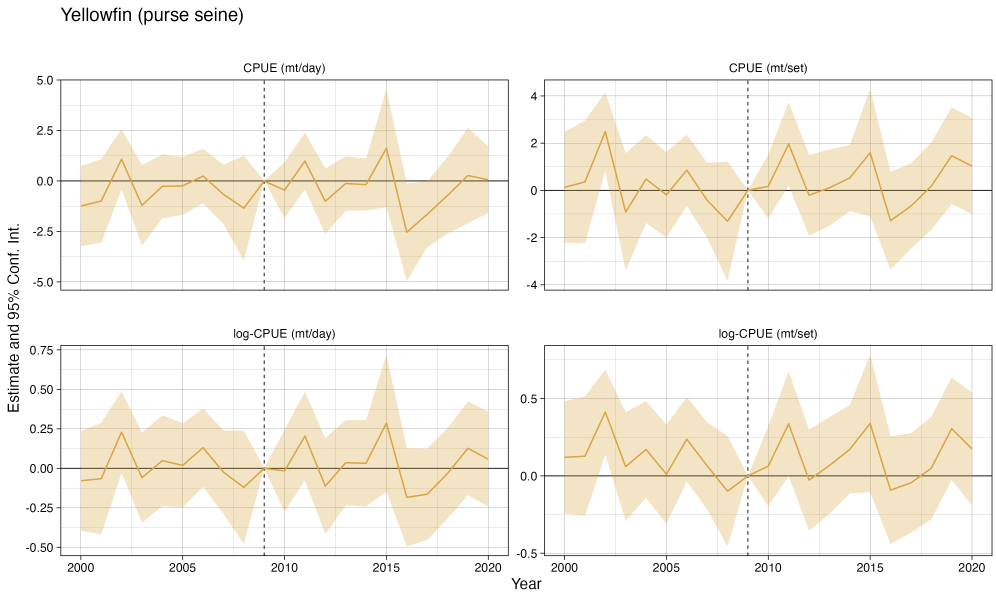



Figure B10: Event-study estimates of the change in purse seine catch-per-unit-effort (CPUE) for yellowfin tuna within 100 nm of the high seas pockets, relative to changes in CPUE observed in areas further away (100–200 nm). Both the near (treated) and far (control) samples are restricted to grid cells within PNA member Exclusive Economic Zones. Estimates are shown for CPUE measured in tons per day (left panels) and tons per set (right panels), in levels (top panels) and for log-transformed CPUE (bottom panels).

Table B10: Coefficient estimates for the change in yellowfin tuna CPUE in the purse seine fleet in areas within 100 nautical miles of the high seas pockets after the closure, relative to the change in CPUE observed for areas between 100 and 200 nautical miles and inside PNA Exclusive Economic Zones. Panel A presents results in levels. Panel B presents results in which the dependent variable is log-transformed.

| | CPUE (mt/day) | CPUE (mt/set) |
|---|---|---|
| *A) Levels* | | |
| Post x Near | 0.146 | 0.402 |
| | (0.281) | (0.300) |
| Num.Obs. | 5956 | 5956 |
| $R^2$ Adj. | 0.221 | 0.280 |
| *B) Log-transformed* | | |
| Post x Near | 0.015 | 0.022 |
| | (0.037) | (0.044) |
| Num.Obs. | 5956 | 5956 |
| $R^2$ Adj. | 0.244 | 0.304 |

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

The unit of observation is a grid cell in a given year. All model specifications include fixed effects by year and grid cell. Numbers in parentheses are Conley standard errors with a 220km radius.

**Longline.** There are only 3 cells within 100–200 nautical miles, leaving us with a reduced control group for this portion of the analyses. These results are presented to comply with our preregistration, but we note the lack of robustness that comes with such a limited sample size. With this caveat in mind, we detect no significant spillover effects for all species combined (Table B11; Figure B11), bigeye tuna (Table B13; Figure B13), or yellowfin tuna (Table B14; Figure B14). The only marginally significant result is for albacore tuna under the log-transformed specification ($p < 0.1$; Table B12; Figure B12).

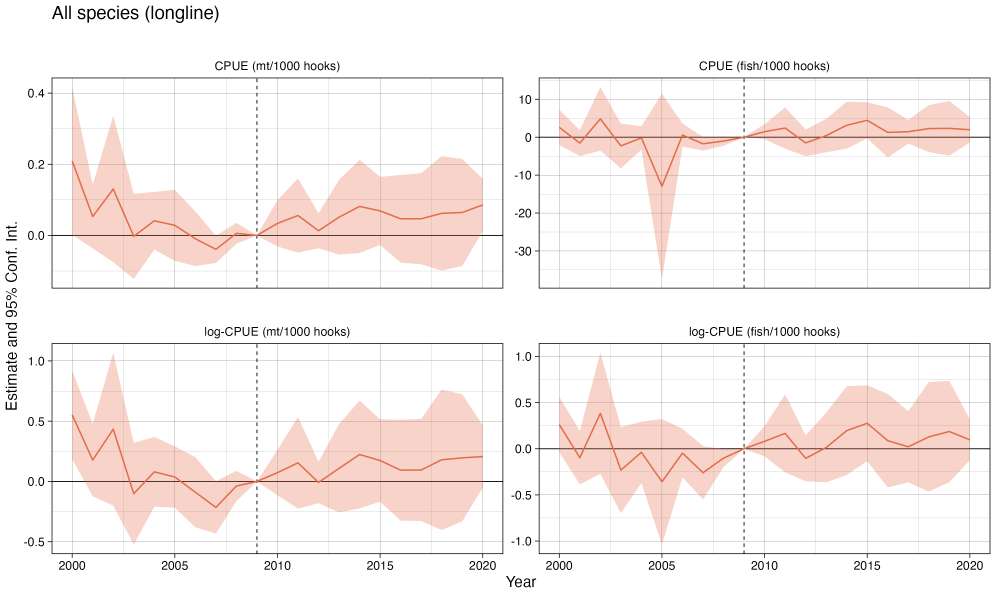



Figure B11: Event-study estimates of the change in longline catch-per-unit-effort (CPUE) across all species combined within 100 nm of the high seas pockets, relative to changes in CPUE observed in areas further away (100–200 nm). Estimates are shown for CPUE measured in metric tons per 1000 hooks (left panels) and number of fish per 1000 hooks (right panels), in levels (top panels) and for log-transformed CPUE (bottom panels).

Table B11: Coefficient estimates for the change in all tuna CPUE in the longline fleet in areas within 100 nautical miles of the high seas pockets after the closure, relative to the change in CPUE observed for areas between 100 and 200 nautical miles. Panel A presents results in levels. Panel B presents results in which the dependent variable is log-transformed.

| | CPUE (fish/1000 hooks) | CPUE (mt/1000 hooks) |
|---|---|---|
| *A) Levels* | | |
| Post x Near | 2.983 | 0.015 |
| | (2.922) | (0.039) |
| Num.Obs. | 347 | 347 |
| $R^2$ Adj. | 0.578 | 0.360 |
| *B) Log-transformed* | | |
| Post x Near | 0.145 | 0.057 |
| | (0.175) | (0.128) |
| Num.Obs. | 347 | 347 |
| $R^2$ Adj. | 0.662 | 0.404 |

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

The unit of observation is a grid cell in a given year. All model specifications include fixed effects by year and grid cell. Numbers in parentheses are Conley standard errors with a 220km radius.

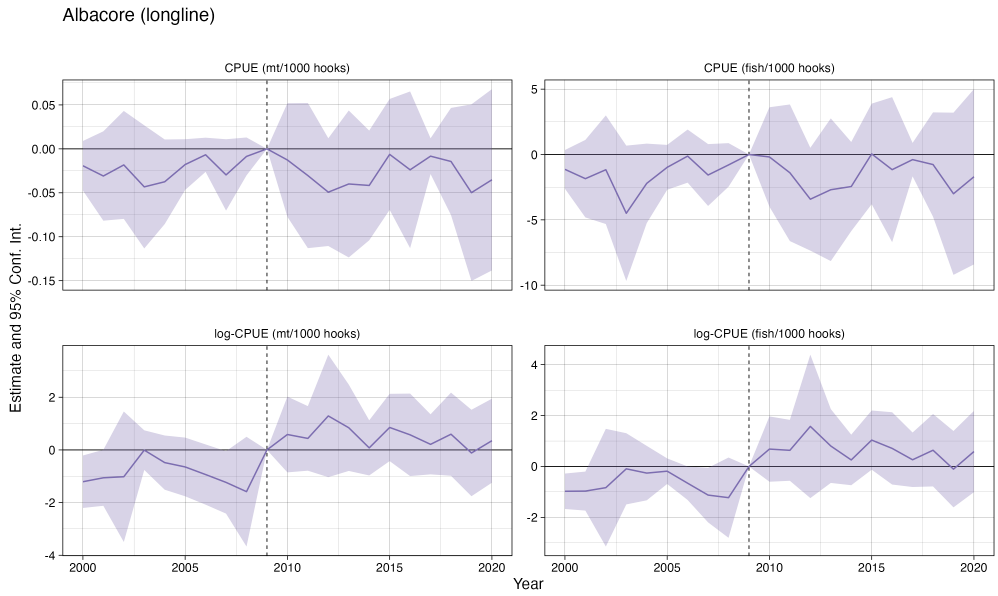



Figure B12: Event-study estimates of the change in longline catch-per-unit-effort (CPUE) for albacore tuna within 100 nm of the high seas pockets, relative to changes in CPUE observed in areas further away (100–200 nm). Estimates are shown for CPUE measured in metric tons per 1000 hooks (left panels) and number of fish per 1000 hooks (right panels), in levels (top panels) and for log-transformed CPUE (bottom panels).

Table B12: Coefficient estimates for the change in albacore tuna CPUE in the longline fleet in areas within 100 nautical miles of the high seas pockets after the closure, relative to the change in CPUE observed for areas between 100 and 200 nautical miles. Panel A presents results in levels. Panel B presents results in which the dependent variable is log-transformed.

| | CPUE (fish/1000 hooks) | CPUE (mt/1000 hooks) |
|---|---|---|
| *A) Levels* | | |
| Post x Near | -0.487 | -0.011 |
| | (1.262) | (0.019) |
| Num.Obs. | 339 | 339 |
| $R^2$ Adj. | 0.878 | 0.881 |
| *B) Log-transformed* | | |
| Post x Near | 1.326* | 1.403* |
| | (0.726) | (0.811) |
| Num.Obs. | 339 | 339 |
| $R^2$ Adj. | 0.843 | 0.819 |

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

The unit of observation is a grid cell in a given year. All model specifications include fixed effects by year and grid cell. Numbers in parentheses are Conley standard errors with a 220km radius.

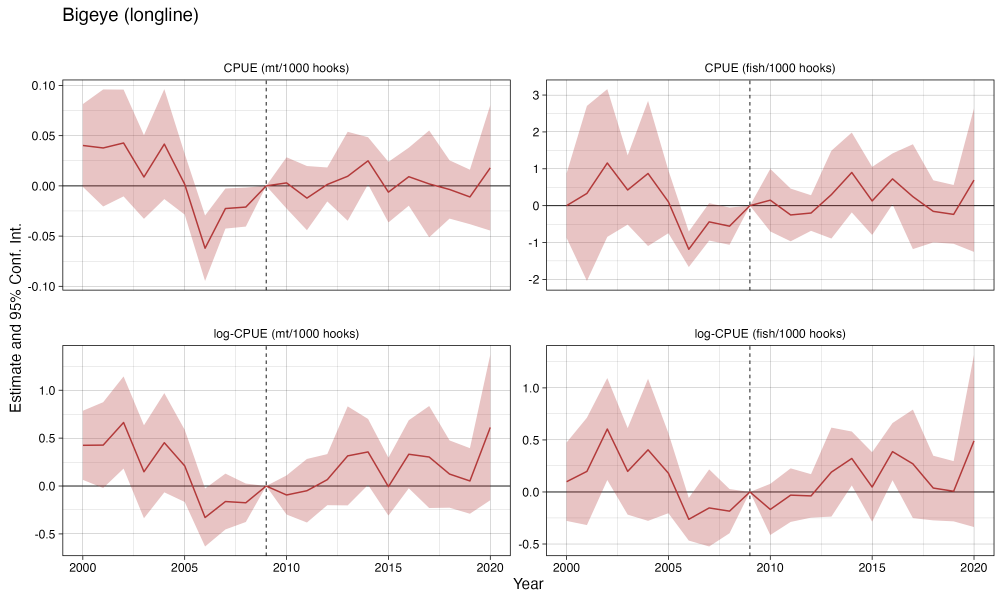



Figure B13: Event-study estimates of the change in longline catch-per-unit-effort (CPUE) for bigeye tuna within 100 nm of the high seas pockets, relative to changes in CPUE observed in areas further away (100–200 nm). Estimates are shown for CPUE measured in metric tons per 1000 hooks (left panels) and number of fish per 1000 hooks (right panels), in levels (top panels) and for log-transformed CPUE (bottom panels).

Table B13: Coefficient estimates for the change in bigeye tuna CPUE in the longline fleet in areas within 100 nautical miles of the high seas pockets after the closure, relative to the change in CPUE observed for areas between 100 and 200 nautical miles. Panel A presents results in levels. Panel B presents results in which the dependent variable is log-transformed.

| | CPUE (fish/1000 hooks) | CPUE (mt/1000 hooks) |
|---|---|---|
| *A) Levels* | | |
| Post x Near | 0.136 | -0.000 |
| | (0.434) | (0.013) |
| Num.Obs. | 347 | 347 |
| $R^2$ Adj. | 0.683 | 0.709 |
| *B) Log-transformed* | | |
| Post x Near | 0.028 | 0.032 |
| | (0.134) | (0.152) |
| Num.Obs. | 347 | 347 |
| $R^2$ Adj. | 0.812 | 0.848 |

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

The unit of observation is a grid cell in a given year. All model specifications include fixed effects by year and grid cell. Numbers in parentheses are Conley standard errors with a 220km radius.




Figure B14: Event-study estimates of the change in longline catch-per-unit-effort (CPUE) for yellowfin tuna within 100 nm of the high seas pockets, relative to changes in CPUE observed in areas further away (100–200 nm). Estimates are shown for CPUE measured in metric tons per 1000 hooks (left panels) and number of fish per 1000 hooks (right panels), in levels (top panels) and for log-transformed CPUE (bottom panels).

Table B14: Coefficient estimates for the change in yellowfin tuna CPUE in the longline fleet in areas within 100 nautical miles of the high seas pockets after the closure, relative to the change in CPUE observed for areas between 100 and 200 nautical miles. Panel A presents results in levels. Panel B presents results in which the dependent variable is log-transformed.

| | CPUE (fish/1000 hooks) | CPUE (mt/1000 hooks) |
|---|---|---|
| *A) Levels* | | |
| Post x Near | 3.346 | 0.026 |
| | (2.150) | (0.030) |
| Num.Obs. | 347 | 347 |
| $R^2$ Adj. | 0.383 | 0.382 |
| *B) Log-transformed* | | |
| Post x Near | 0.254 | 0.176 |
| | (0.209) | (0.201) |
| Num.Obs. | 347 | 347 |
| $R^2$ Adj. | 0.518 | 0.493 |

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

The unit of observation is a grid cell in a given year. All model specifications include fixed effects by year and grid cell. Numbers in parentheses are Conley standard errors with a 220km radius.

# C Robustness Tests

## C.1 Results by high seas pocket

To assess whether our main results are driven by a single pocket, and as a robustness test to account for the exemption granted to the Philippines to fish in the Western pocket (HSP1), we re-estimate the main specifications testing for change in bigeye tuna longline CPUE within the pockets (hypothesis 2) and for spillover benefits to tuna purse seine with CPUE of skipjack (hypohtesis 3) separately for each pocket. For the first analysis, we drop treated grid cells from one pocket at a time while retaining the full set of control cells. For the spillover analysis, we split the entire sample—both near and far grid cells—east and west of 152.5°. The increase in bigeye CPUE measured in number of fish is positive and significant for both the western ($p < 0.01$) and eastern ($p < 0.05$) pockets, with a larger point estimate in the western pocket (Table C1). The null spillover result for skipjack tuna holds for both pockets individually (Table C2).

Table C1: Coefficient estimates for change in bigeye tuna CPUE caught by the longline fleet in the high seas pockets after the closure, relative to changes in CPUE observed for other tropical (20°S - 20°N) high seas areas in the WCPFC convention area. Models are estimated separately for HSP1 (western pocket, lon $\leq$ 152.5°) and HSP2 (eastern pocket, lon $>$ 152.5°; See Figure 1). The first column shows our main text estimates as in Table 2.

| | sample: Full sample | sample: HSP1 | sample: HSP2 |
|---|---|---|---|
| *A) CPUE (fish / 1000 hooks)* | | | |
| Post x Treated | 0.697* | 1.537*** | 0.397** |
| | (0.363) | (0.520) | (0.196) |
| Num.Obs. | 1631 | 1407 | 1529 |
| $R^2$ Adj. | 0.798 | 0.801 | 0.804 |
| *B) CPUE (mt / 1000 hooks)* | | | |
| Post x Treated | -0.003 | 0.020 | -0.011 |
| | (0.011) | (0.013) | (0.010) |
| Num.Obs. | 1631 | 1407 | 1529 |
| $R^2$ Adj. | 0.829 | 0.830 | 0.835 |

* p $<$ 0.1, ** p $<$ 0.05, *** p $<$ 0.01

The unit of observation is a grid cell in a given year. All model specifications include fixed effects by year and grid cell. Numbers in parentheses are Conley standard errors with a 1100km radius.

Table C2: Coefficient estimates for change in skipjack tuna CPUE caught by the purse seine fleet in areas within 100 nautical miles of each high seas pocket after the closure. Models are estimated separately for HSP1 (western pocket, lon $\leq 152.5°$) and HSP2 (eastern pocket, lon $> 152.5°$); See Figure 1. The first column shows our main text estimates as in Table 3.

| | sample: Full sample | sample: HSP1 | sample: HSP2 |
|---|---|---|---|
| *A) CPUE (mt/day)* | | | |
| Post x Near | -0.496 | -0.132 | -0.952 |
| | (0.680) | (0.635) | (0.957) |
| Num.Obs. | 5960 | 1645 | 4315 |
| $R^2$ Adj. | 0.333 | 0.433 | 0.319 |
| *B) CPUE (mt/set)* | | | |
| Post x Near | -0.865 | -0.925 | -1.206 |
| | (0.818) | (0.716) | (1.141) |
| Num.Obs. | 5960 | 1645 | 4315 |
| $R^2$ Adj. | 0.245 | 0.353 | 0.213 |

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

The unit of observation is a grid cell in a given year. All model specifications include fixed effects by year and grid cell. Numbers in parentheses are Conley standard errors with a 220km radius.